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Thermal-hydraulic characteristics analysis of unprotected accident and protection control strategy for helium-xenon cooled reactor system

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ABSTRACT

The small modular helium-xenon cooled reactor has the advantages of simple and compact system structure, short construction cycle, light weight and small volume, which is very suitable for energy supply in remote area and small nuclear propulsion device. The research on accident characteristics and protection control strategies is very important to ensure the safe and reliable operation of the reactor system and mitigate the consequences of the accident. Therefore, the safety characteristics analysis program of helium-xenon cooled reactor system is developed based on Modelica language, and the transient verification of typical working conditions is carried out. The maximum relative error of the verification results was 7.91 %. Through the program, the unprotected transient conditions and corresponding protection control strategy transient conditions of external load loss accident, partial loss of heat sink accident, main valve accidental closing accident and Brayton unit efficiency step decline accident are simulated. The results show that the loss of all external loads will lead to rotation speed overspeed. The introduction of negative reactivity or bypass control can effectively control the rotation speed rise, but the introduction of negative reactivity is better than bypass control. Under unprotected control, the partial loss of the heat sink, the accidental closing of the main valve and the step down of the Brayton unit efficiency will all lead to the rapid decline of the rotation speed and mass flow rate, and there is a risk of the reactor temperature exceeding the limit. The rotation speed and mass flow rate can be effectively stabilized by load following to prevent the temperature exceeding the limit.

Nomenclature

Symbols			
G	Mass flow rate (kg/s)	*in*	Inlet
P	Pressure (Pa)	*out*	Outlet
ΔP	Pressure drop (Pa)	*av*	Average
T	Temperature (K)	*f*	Friction
ΔT	Temperature change relative to rated condition (K)	*a*	Local
		ac	Accelerate
W	Power (W)	*z*	Gravitational
M	Quality (kg)	*mix*	Helium-xenon mixture

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H	Specific enthalpy (kJ/kg)	*h*	Helium
V	Speed (m/s)	*x*	Xenon
Z	Vertical position (m)	*fl*	Nuclear fuel
C_p	Constant pressure specific heat (J/(kg·K))	*s*	Structural material
Cv	Constant volume specific heat (J/(kg·K))	*fn*	Reactor fission
R_g	Gas constant of He–Xe mixture gas (J/(kg·K))	*he*	Reactor heat exchange
θ	Heat transfer resistance (K/W)	*fc*	Between coolant and nuclear fuel

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μ	Dynamic viscosity (Pa·s)	*sc*	Between coolant and structural material
Ψ	Experimental data fitting function	*is*	Outside introduction
m	Molar mass (kg/mol)	*cl*	Coolant
λ	Thermal conductivity (W/(m·K))	*e*	Heat exchanger
X	Mass fraction in He–Xe mixture gas	*ho*	Hot side coolant
Λ	Reactivity	*co*	Cold side coolant
δ	Feedback coefficient (K^{-1})	*tu*	Turbine
B	Second virial coefficient	*c*	Compressor
C	Third virial coefficient	**Superscripts**	
Y	Actual gas compression coefficient	0	Weighted average
ρ	Density (kg/m^3)	*	Fitted
g	Gravitational acceleration (m/s^2)	**Abbreviations**	
l	Length (M)	MLD	Main logic diagram
A	Flow area (m^2)	PSA	Probabilistic safety assessment
t	Time (s)	SMR	Small modular reactor
L	Specific power (J/kg)	CFD	Computational fluid dynamics
η	Efficiency	SCO_2	Supercritical carbon dioxide
v	Average critical specific volume (m^3/kg)	JIMO	Jupiter icy moons orbiter
Subscripts		Nu	Nusselt number

1. Introduction

With the development of the era, the speed of global energy consumption has accelerated, and the traditional fossil energy has been gradually depleted [1–3]. Nuclear energy is a new type of clean and low-carbon energy, and there is still much room for development and utilization [4–6]. The development and utilization of nuclear energy are of great significance for achieving sustainable development and ensuring energy security [7–9]. For application scenarios such as energy supply in remote area and small nuclear propulsion device, currently more mature pressurized water reactor or boiling water reactor are not suitable for use due to their large volume, complex system, and long construction cycle. However, small modular reactor (SMR) have a simple and compact system structure, short construction period, light weight, and small volume, which are very suitable for the above application scenarios and have attracted the research interest of many researchers in the field of nuclear energy [10–15]. At present, there are four types of SMR: micro heat pipe cooled reactor, helium cooled reactor, helium-xenon cooled reactor and SCO_2 reactor. The power of micro heat pipe cooled reactor is generally kilowatt level [16], which is insufficient for the above application scenarios. Helium cooled reactor, helium-xenon cooled reactor and SCO_2 reactor are all direct closed Brayton cycle reactor with compact and high-power characteristics [17–22]. The size of turbine and compressor of helium cooled reactor system is large [23], which is not suitable for nuclear power plant in small space. SCO_2 is more corrosive than helium at high temperature [24], and the operating pressure of the SCO_2 reactor system is high [25], which has problems of leakage and pressure bearing. However, the helium-xenon mixed gas not only contains the excellent inert and heat transfer ability of helium, but also the turbine and compressor size of helium-xenon cooled reactor system is much smaller than that of helium cooled reactor system [26]. Therefore, from the perspective of size and safety, SMR is more suitable for helium-xenon cooled reactor.

Since 1968, NASA has carried out research on Brayton cycle system using helium-xenon gas mixture as working medium [27], and helium-xenon Brayton cycle has begun to enter the vision of researchers. In 1986, NASA planned to build a solar helium-xenon Brayton system with turbine inlet temperature as high as 1034k [28]. In 2000, the first helium-xenon Brayton cycle nuclear power system appeared with NASA's Prometheus program and the Jupiter Icy Moons Orbiter (JIMO) [29,30]. Since then, the conceptual scheme and basic principle of Brayton cycle system for helium-xenon cooled reactor have been preliminarily formed.

In terms of the research progress of helium-xenon Brayton cycle and helium-xenon cooled reactor, El Genk et al. [26,31] analyzed the applicability of helium-xenon mixed gas and helium for closed Brayton cycle, and proposed that the helium-xenon mixed ratio of 40 g/mol and 15 g/mol has better heat transfer capacity and smaller turbine size, which is more suitable for the space power system of nuclear reactor, but the pressure loss of helium-xenon circuit is higher than that of helium. Yang et al. [32] analyzed the influence of different helium-xenon mixing ratios on the adiabatic coefficient, regenerator efficiency and pressure loss by establishing the thermodynamic analysis model of helium-xenon Brayton cycle. The adiabatic coefficient and pressure loss increased with the increase of molar weight, and the regenerator efficiency first increased and then decreased with the increase of molar weight. Zhang et al. [33] developed a transient analysis program by establishing the dynamic models of various components of the helium-xenon Brayton reactor system, simulated the transient process of a single Brayton cycle failure shutdown accident and reactivity introduction accident, and proposed that the helium-xenon reactor system with double circuits is safer. Wang et al. [34] developed transient analysis code by establishing transient model of key equipment of helium-xenon reactor space power system, simulated transient conditions of positive reactivity introduction, system shutdown and residual heat removal, and concluded that a certain amount of reactivity introduction can be adjusted and stabilized by negative feedback, and residual heat can be removed in a passive manner. Li et al. [35] developed a three-dimensional simulation platform of helium-xenon cooled reactor based on OpenMC, Dealii and Fenics, and verified the reliability of the platform. Wang et al. [36] studied the influence of various factors on *Nu* through CFD three-dimensional numerical simulation, and established and verified the semi empirical relationship of axial *Nu* distribution. Wang et al. [37] developed the natural circulation system analysis program of helium-xenon cooled reactor, and analyzed the residual heat removal characteristics of helium-xenon natural circulation through the program. Zhou et al. [38]preliminarily sorted out the start-up events of helium-xenon cooled reactor through MLD method, laying the foundation for PSA analysis. Ma et al. [39] carried out research on thermal performance optimization and quality evaluation of helium-xenon cooled reactor Brayton system, studied the relationship between system efficiency and quality, and concluded that the reactor Brayton cycle system using helium-xenon mixed as cycle working fluid was the optimal solution in terms of thermal performance and quality through optimization design. In addition, there are also some research results on the reactor Brayton cycle system using other cycle working fluids. Chen et al. [40] took the reactor Brayton system using air as the cycle working fluid as the research object, designed the system parameters and optimized the system parameters from three aspects of power density, volume and efficiency through genetic algorithm. Li et al. [41] took the Brayton cycle system of heat pipe reactor as the research object, analyzed the load following characteristics of the system, and proposed the high efficiency operation curve of the system. Li et al. [42] took the Brayton cycle system using SCO_2 as the cycle working fluid as the research object, carried out the system start-up experimental research, and proposed the corresponding start-up control strategy and the problems needing attention in the start-up process.

The study of accident characteristics and protection control strategies of helium-xenon cooled reactor system is very important to ensure the safe and reliable operation of the system and mitigate the consequences of the accident. At present, the research on helium-xenon Brayton cycle and helium-xenon cooled reactor mainly focuses on the influence of helium-xenon gas mixture on Brayton cycle, the flow heat

transfer and nuclear physical characteristics of helium-xenon cooled reactor, and the residual heat removal system. For the Brayton cycle system with various cycle working fluids, the previous research results mainly focused on the design optimization of system and the study of system startup characteristics. In the aspect of system safety characteristics research, previous studies mainly focused on the characteristics of reactivity introduction, a common typical accident. There are still many deficiencies in the research of system safety characteristics, and there is a lack of research on safety characteristics and protection control strategies under more accident scenarios. Therefore, it is necessary to study the safety characteristics of unprotected accident and protection control strategies for helium-xenon cooled reactor system. The innovations and key scientific problems to be solved in this paper are as follows: (1) The unprotected safety characteristics of helium-xenon cooled reactor system under various accidents are studied to provide parameter basis for accident identification and support the research of protection control strategy under corresponding accident conditions; (2) Study the protection and control strategies under various accidents, provide reliable accident protection and control strategies, and find the optimal strategy, so as to provide guidance for the protection and control strategies that should be adopted in different accidents.

In this paper, the safety characteristics analysis program of helium-xenon cooled reactor system is developed based on Modelica, and the correctness and reliability of the program are preliminarily verified by transient comparison with references. The unprotected transient process of helium-xenon cooled reactor system under the conditions of external load loss accident, partial loss of heat sink accident, main valve accidental closure accident and Brayton unit efficiency step decrease accident was simulated, and their transient characteristics were analyzed. According to the unprotected transient characteristics of each other accident, the corresponding protection control strategies is proposed, and the accident transient process after the protection control is simulated, and the safety benefits brought by the protection control strategies are analyzed.

2. Research content and method

2.1. Research object - system description

The research object of this paper is the helium-xenon cold reactor system with direct Brayton cycle. The working fluid in Brayton cycle circuit is 40 g/mol helium-xenon mixed gas. The Brayton cycle system uses external water source as the final heat sink for the discharge of waste heat. Fig. 1 depicts the schematic diagram of helium-xenon cold reactor system, which is composed of Brayton cycle main system, volume charging and discharging auxiliary protection system and bypass control auxiliary protection system.

For the Brayton cycle main system, the main equipment is reactor, turbine, compressor, regenerator and cooler. As a heat source, reactor provides basic heat energy for Brayton cycle. The turbine depressurizes and cools the flowing working medium and does work to provide output power for the compressor and external loads. The function of the regenerator is to improve the circulating heat efficiency through heat recovery and utilization. The compressor pressurizes and warms the flowing working medium to provide the high pressure required for turbine expansion and the driving pressure required for working medium circulation. According to the second law of thermodynamics, the thermal cycle needs to discharge a large amount of waste heat to do work, and the waste heat of Brayton cycle is discharged through the cooler.

For the volume charging and discharging auxiliary protection system, the main equipment is the volume tank, volume discharging valve and volume charging valve. The initial pressure in the volume box is the average of the inlet and outlet pressures of the compressor. The pressure difference drives the working medium to charge from the volume box to the inlet of the compressor or discharge from the outlet of the compressor to the volume box. The volume discharge valve and volume charge valve realize the control of volume charge and discharge. The auxiliary protection system for bypass control is composed of bypass

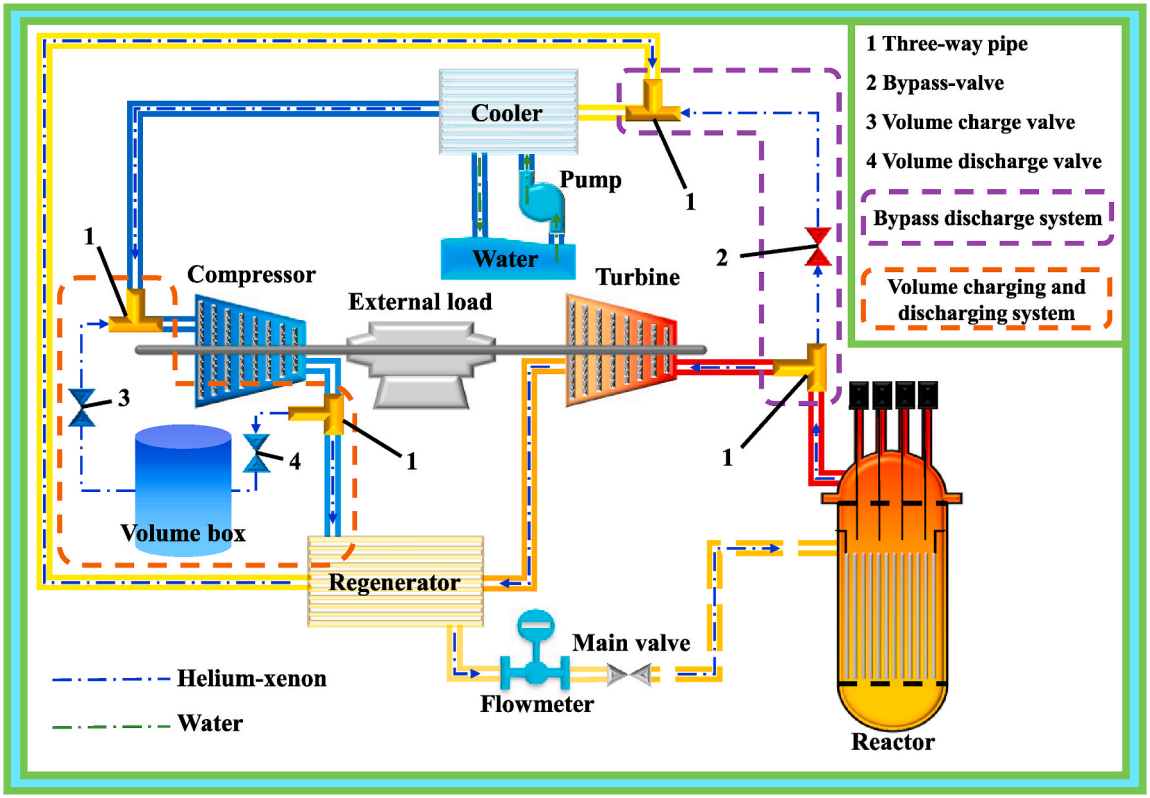


Fig. 1. Schematic diagram of helium-xenon cooled Brayton reactor system and protection control system.

valve and its pipeline, and the bypass valve realizes the start and stop of bypass discharge. When the bypass valve is opened, the working fluid is driven by the pressure difference between the reactor outlet and the cooler inlet, and flows directly from the reactor outlet to the cooler through the bypass pipeline, reducing the turbine mass flow rate.

2.2. Research method

The research method of safety characteristics of helium-xenon cooled reactor system is to develop the safety characteristics analysis program of helium-xenon cooled reactor system through Modelica language, carry out the transient simulation of accidents under unprotected control and protected control, and analyze the corresponding transient characteristics. At present, there have been many researches on the modeling of system analysis program, and the corresponding mathematical equation model has been relatively common and mature [11,13,22,34,43–45]. Therefore, this paper will not describe each model in detail. Only some important core equations are described in the paper. For example, the mass conservation, momentum conservation and energy conservation equations that each equipment model needs to meet, the physical property calculation equation of helium-xenon gas mixture, the key mathematical equations of the reactor and the mathematical equations required for theoretical analysis. The program modeling interface is shown in Fig. 2. The system safety characteristic analysis program is developed based on the mass conservation equation, momentum conservation equation, energy conservation equation and gas state equation, and the corresponding mathematical models are Eq. (1) to Eq. (4) [13,45–48].

$$G_{out} - G_{in} = \frac{dM}{dt} \tag{1}$$

$$P_{in} - P_{out} = \frac{l}{A}\frac{dG_{av}}{dt} + \Delta P_f + \Delta P_a + \Delta P_v + \Delta P_z \tag{2}$$

$$W = G_{out}\left(H_{out} + \frac{V_{out}^2}{2} + Z_{out}g\right) - G_{in}\left(H_{in} + \frac{V_{in}^2}{2} + Z_{in}g\right) + MC_p\frac{d(T_{av})}{dt} \tag{3}$$

$$\rho = \frac{P}{YR_gT} \tag{4}$$

where, G_{out} is the outlet mass flow rate; G_{in} is the inlet mass flow rate; M is the quality of coolant inside the equipment; t is the time; P_{out} is the outlet pressure; P_{in} is the inlet pressure; l is the length; A is the flow area; G_{av} is the average mass flow rate; ΔP_f is the friction pressure drop; ΔP_a is the local pressure drop; ΔP_v is the accelerated pressure drop; ΔP_z is the gravitational pressure drop; W is the power; H_{out} is the outlet specific enthalpy; H_{in} is the inlet specific enthalpy; V_{out} is the outlet speed; V_{in} is the inlet speed; Z_{out} is the outlet vertical position; Z_{in} is the inlet vertical position; C_p is the specific heat at constant pressure; T_{av} is the coolant average temperature; g is the gravitational acceleration; ρ is the density; T is the temperature; P is the pressure; R_g is the gas constant of helium-xenon mixture gas; Y is the actual gas compression coefficient.

In the safety characteristics analysis program of helium-xenon cooled reactor system, the main physical parameters of helium-xenon gas mixture are density, dynamic viscosity, specific heat at constant pressure and thermal conductivity. The mathematical model of density is Eq. (4), and the mathematical model of other physical parameters is Eq. (5) to Eq. (9) [45].

$$Y = 1 + B\rho + C\rho^2 \tag{5}$$

$$\mu = \mu^0 + \left(1 - \frac{1}{2.3}\right)\mu^* \times \psi_u\left(\frac{0.291 \times \nu^*}{m_{mix}}\rho\right) \tag{6}$$

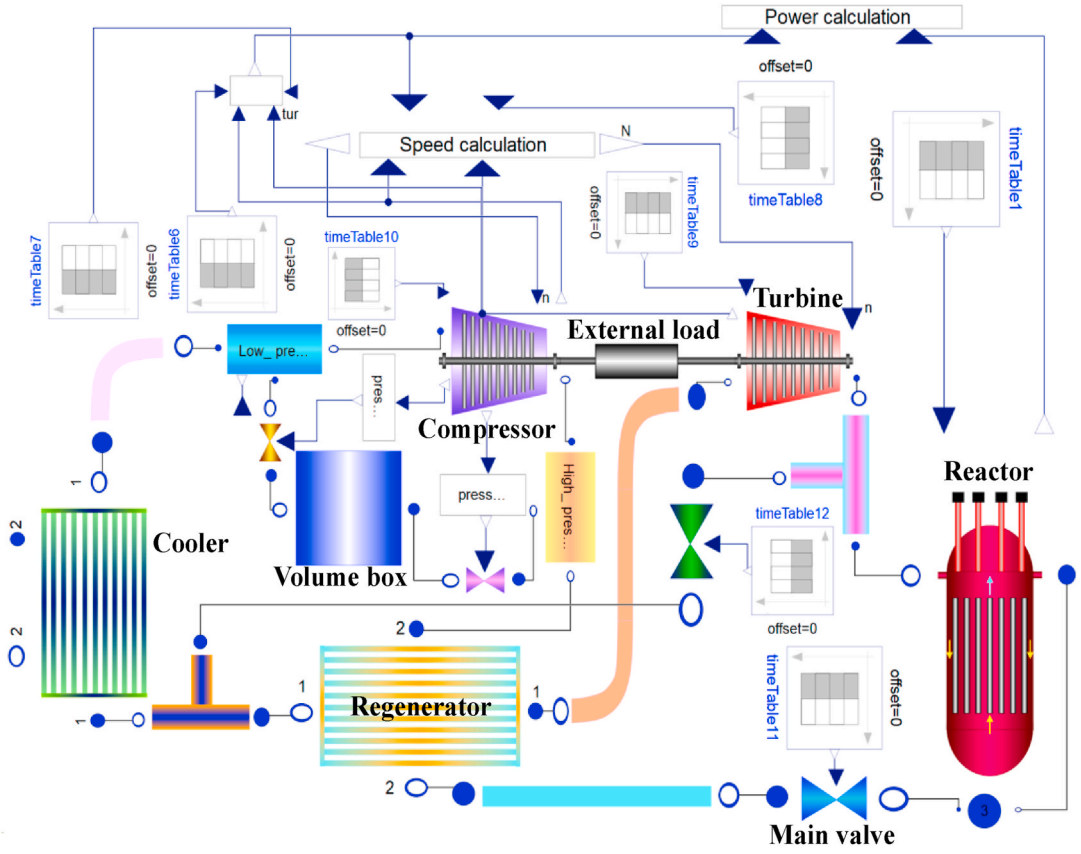


Fig. 2. Model diagram of accident safety characteristics analysis program for helium-xenon cooled Brayton reactor system.

$$\lambda = \lambda^0 + \left(1 - \frac{1}{2.94}\right)\lambda^* \times \psi_\lambda\left(\frac{0.291 \times v^*}{m_{mix}}\rho\right) \tag{7}$$

$$m_{mix} = X_h m_h + X_x m_x \tag{8}$$

$$C_p = 2.6276{X_h}^5 - 5.3238{X_h}^4 + 4.1073{X_h}^3 - 1.4134{X_h}^2 + 0.2345X_h + 0.0071 \tag{9}$$

where, B is the second virial coefficient; C is the third virial coefficient; μ is the dynamic viscosity; μ^0 is the weighted average of dynamic viscosity; μ^* is the fitting dynamic viscosity; v^* is the average critical specific volume; $\Psi_\mu(\rho)$ is the dynamic viscosity fitting function of test data; m_{mix} is the molar mass of helium-xenon mixture; λ is the thermal conductivity; λ^0 is the weighted average of thermal conductivity; λ^* is the fitting thermal conductivity; $\Psi_\lambda(\rho)$ is the thermal conductivity fitting function of test data; X_h is the mass fraction of helium in helium-xenon mixture; X_x is the mass fraction of xenon in helium-xenon mixture; m_h is the molar mass of helium; m_x is the molar mass of xenon.

As shown in Fig. 3, the pressure and mass flow rate are coupled by the momentum conservation equation, the pressure and temperature are coupled by the gas state equation, and the mass flow and temperature are coupled by the energy conservation equation and equipment power. Therefore, the system pressure, temperature, and mass flow rate form a closed-loop coupled iterative solution. The reactor power is divided into two types: fission power and heat transfer power. When solving the energy conservation Eq. (3), the power is the reactor heat exchange power. The reactor heat exchange power is determined by Eq. (11), The relationship between reactor heat exchange and fission power is shown in Eq. (10). The fission power of the reactor is solved by the total reactivity in Eq. (12) and the point reactor neutron kinetics equation. The point reactor neutron kinetics equation is widely used and will not be described in this paper. The heat exchange power of the heat exchanger is calculated by the temperature difference of the coolant on both cold side and hot side through Eq. (13) [13,44]. The power of compressor and turbine is volume power, which is solved by their density and the pressure difference of compression or expansion. The specific mathematical model will be described in section 4.1, as shown in Eq. (14) to Eq. (17) [49].

$$M_{fl}Cv_{fl}\frac{dT_{fl}}{dt} + M_sCv_s\frac{dT_s}{dt} = W_{fn} - W_{he} \tag{10}$$

$$W_{he} = \frac{(T_{fl} - T_{av})}{\theta_{fc}} = \frac{(T_s - T_{av})}{\theta_{sc}} \tag{11}$$

$$\Lambda = \Lambda_{is} + \delta_{cl}\Delta T_{av} + \delta_s\Delta T_s + \delta_{fl}\Delta T_{fl} \tag{12}$$

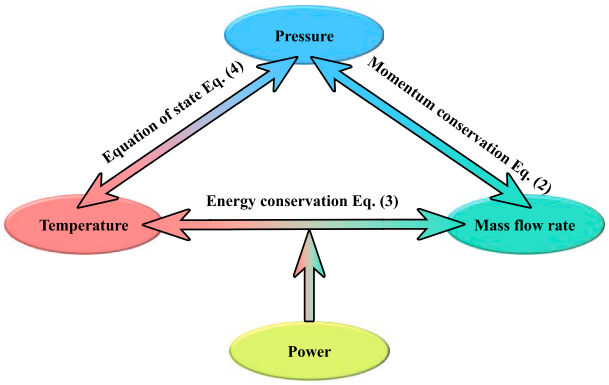


Fig. 3. General solution idea of mathematical model of system analysis program.

$$W_e = \frac{T_{ho} - T_{co}}{\theta_e} \tag{13}$$

where, M_{fl} is the nuclear fuel quality; Cv_{fl} is the constant volume specific heat of nuclear fuel; T_{fl} is the average temperature of nuclear fuel; M_s is the structural material quality; Cv_s is the constant volume specific heat of structural material; T_s is the average temperature of structural material; W_{fn} is the reactor fission power; W_{he} is the reactor heat exchange power; θ_{fc} is the heat transfer resistance between coolant and nuclear fuel; θ_{sc} is the heat transfer resistance between coolant and structural material; Λ is the total reactivity; Λ_{is} is the reactivity introduced from outside; δ_{cl} is the coolant feedback coefficient; δ_s is the feedback coefficient of structural material; δ_{fl} is the fuel feedback coefficient; ΔT_{av} is the change in coolant average temperature relative to rated operating condition; ΔT_s is the change in structural material average temperature relative to rated operating condition; ΔT_{fl} is the change in nuclear fuel average temperature relative to rated operating condition; W_e is the heat exchange power of heat exchanger; T_{he} is the average temperature of hot side coolant; T_{co} is the average temperature of cold side coolant; θ_e is the heat transfer resistance between hot side and cold side coolant.

2.3. Protection control strategies and accident sequence

This paper studies the unprotected transient characteristics of helium-xenon cooled reactor system under the conditions of external load loss accident, partial loss of heat sink accident, main valve accidental closure accident and Brayton unit efficiency step decrease accident. Through the analysis of unprotected characteristics of various accident conditions, the corresponding protection control strategies is proposed, and the transient characteristics of the protection control strategies are analyzed. The accident protection control logic is shown in Fig. 4.

Pressure overrun protection is realized by volume discharge. The pressure limit of HTR-10 high temperature gas cooled reactor is 128 % of the rated steady-state pressure [50–53], In this paper, 2.6 MPa is selected as the maximum pressure limit of helium-xenon cooled reactor system in combination with the rated operating pressure of the system and with reference to the pressure limit of HTR-10. The helium-xenon cooled reactor system has a tendency of pressure rise in some accident operation conditions, which may lead to pressure overrun. Therefore, it is necessary to reduce the system pressure through the volume charging and discharging system to prevent pressure overrun. The response command of the volume discharge valve is shown in Table 1. When the system pressure exceeds 2.5 MPa, the opening of the volume discharge

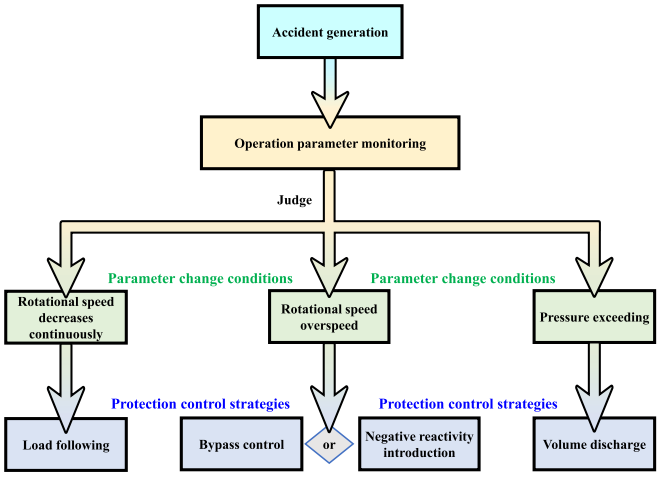


Fig. 4. Protection control strategies adopted to cope with different key parameter changes.

Table 1
Volume discharge valve response command.

Valve type	Pressurizer pressure range	Valve action
Volume discharge valve	>2.6 MPa	Step increase in valve opening to 100 %
Volume discharge valve	>2.5 MPa	Step increase in valve opening to 70 %
Volume discharge valve	<2.1 MPa	Step close the valve

valve rises to 70 % step by step. If the pressure rising trend cannot be suppressed, when the system pressure exceeds the pressure limit of 2.6 MPa, the volume discharge valve will be fully opened for pressure relief. When the system pressure drops to 2.1 MPa due to pressure relief, close the volume discharge valve to prevent pressure and flow loss accident caused by further pressure reduction.

Fig. 5 describes the research ideas of no protection characteristics and protection strategy characteristics of various accidents. For the loss of external load accident, firstly, the unprotected characteristics of the system under the condition of 20 % loss of external load are studied, and then the unprotected characteristics under different degrees of loss of load are studied. Finally, taking all external load loss conditions as an example, the characteristics of bypass control protection and negative reactivity introduction protection are studied according to the rotation speed overspeed characteristic and the protection strategies proposed in Fig. 4. For the main valve accidental closure accident, firstly, the unprotected characteristics of the system under the condition of reducing the opening of the main valve by 20 % were studied, and then the unprotected characteristics under different degrees of reducing the opening of the main valve were studied. Finally, taking the case of reducing the opening of the main valve by 20 % as an example, the characteristics of the load following protection are studied according to the rotation speed continuous decline characteristic and the protection strategy proposed in Fig. 4. For the partial loss of heat sink accident and the efficiency step decline accident of Brayton unit, firstly, the no protection characteristics of the accident are studied, and then the characteristics of load following protection are studied according to the rotation speed continuous decline characteristic and the protection strategy proposed in Fig. 4. In addition, in all protection control simulation cases, when the system pressure exceeds the limit value, overpressure protection will be triggered, and the volume discharge valve will execute Table 1 command.

3. Validation of analysis program

At present, the helium-xenon cooled reactor system still lacks the engineering or experimental data of the whole system, and it is difficult to carry out the experiment of the whole system. In the previous study, we carried out the steady-state verification of the safety characteristics analysis program of the helium-xenon cooled reactor system, and preliminarily verified the accuracy of the system analysis program [54]. In order to further verify the system analysis program, the transient accuracy of the system analysis program is preliminarily verified by comparing the transient simulation results with reference [45]. In the program verification example, the boundary conditions, initial conditions and design parameters of the system are similar to those in the literature [45] as much as possible to better verify the reliability and correctness of the system analysis program. The verification results are shown in Figs. 6 and 7.

As shown in Figs. 6 and 7, for the verification conditions of positive reactivity introduction and negative reactivity introduction, it is rated steady state before 3000s, and positive reactivity and negative reactivity are introduced by step at 3000s. The results calculated by the system analysis program are generally consistent with the literature value in the overall trend, but there is a steady-state initial error because it cannot

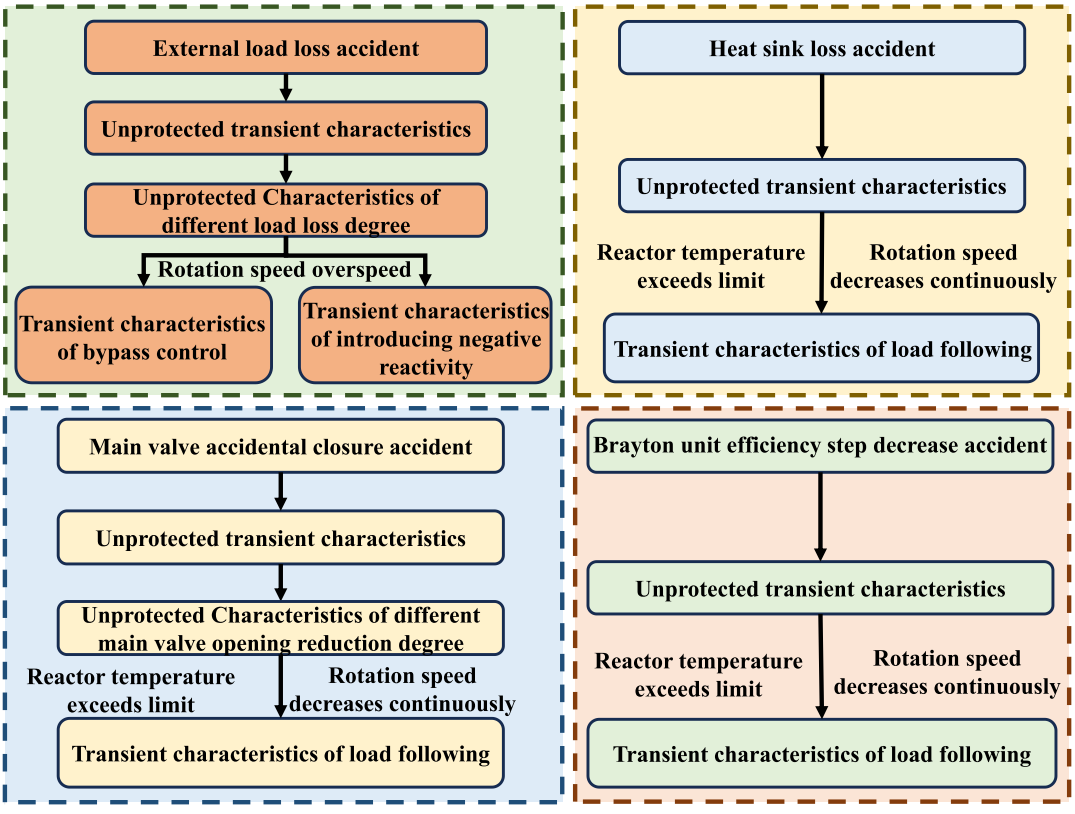


Fig. 5. General idea of research on no protection characteristics and protection control strategies characteristics of system accidents and overview of simulation conditions.

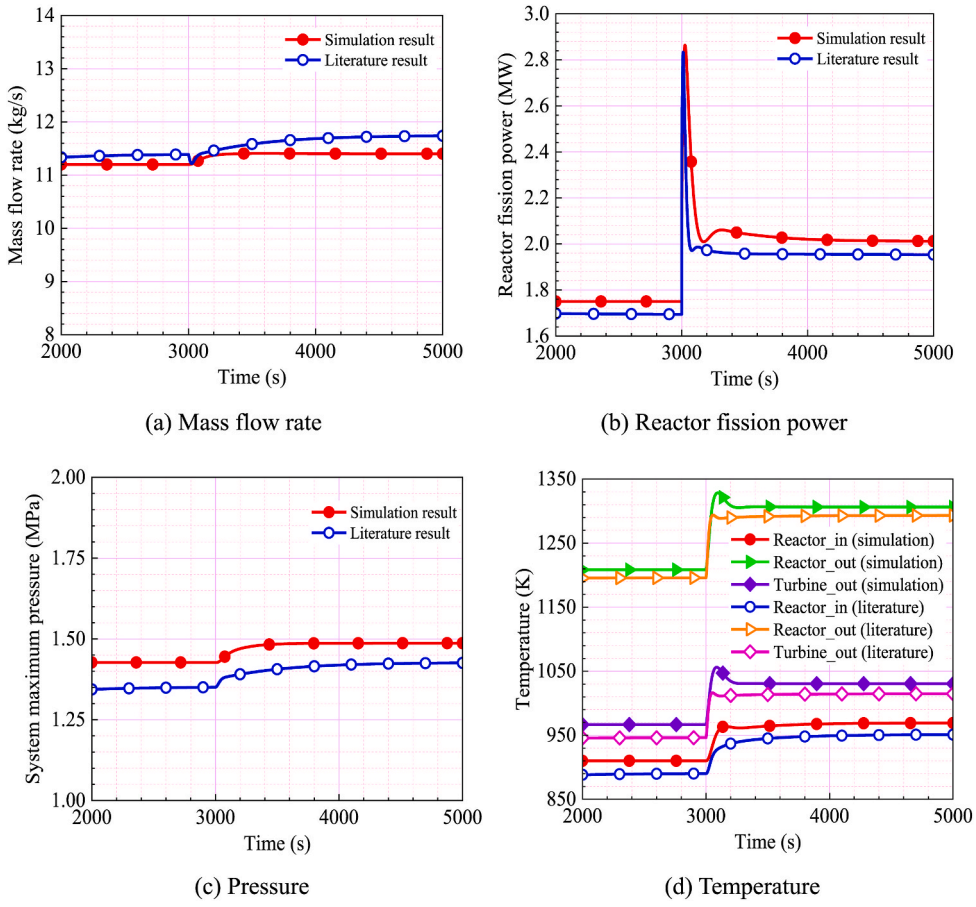


Fig. 6. Verification and comparison of transient curves under the condition of introducing positive reactivity.

reach a very accurate close to the rated steady-state initial value of the system in the literature. In the two verification conditions, the maximum relative error of circulating mass flow rate is 2.87 %, the maximum relative error of reactor fission power is 5.40 %, the maximum relative error of system maximum pressure is 7.91 %, and the maximum relative error of temperature is 4.33 %. The reliability of the safety characteristics analysis program of helium-xenon cooled reactor system is preliminarily verified by transient comparison with reference [45].

4. Results and discussion

The helium-xenon cooled reactor system was in rated steady state operation before the introduction of accident disturbance. The specific steady-state parameters of the system have been listed in the previous research article [54], and will not be repeated in this paper. In addition, for the study of unprotected safety characteristics and protection control strategies of various accidents, the rated steady-state parameters are consistent before the introduction of accident disturbance. In the results and discussion, there are a large number of system parameters transient curves and radar charts for comparative analysis. The overall processing flow of data results is shown in Fig. 8.

4.1. Analysis of unprotected transient characteristics and protection control strategy characteristics under loss of external load accident

The loss of external load accident is one of the typical accidents of nuclear power plant. During the rated steady-state operation of the system, the external load may drop due to the load rejection of Brayton units, resulting in the loss of external load accident. Therefore, this section takes the external load step drop of 20 % as an example to simulate the transient response process of Brayton system after the external load drop, and analyze the transient characteristics. The initial state of the system simulation is the rated steady state condition, and the load power is reduced by 20 % at 500s. The simulation results are shown in Figs. 10 and 11, and the temperature-specific entropy (T-s) diagrams is shown in Fig. 9.

As shown in Figs. 9, 1–2 is the expansion process of the turbine, 2–3 is the cooling process of regenerator, 3–4 is the cooling process of cooler, 4–5 is compression process the of the compressor, 5–6 is the heating process of regenerator, and 6-1 is the heating process of reactor. After the loss of external load accident (500s), with the progress of the accident, the T-s diagram of the reactor system gradually widens and shortens, the heating power of the reactor and the cooling power of the cooler gradually increase, the power lost by the compressor and turbine due to the entropy increase also gradually increases, and the efficiency of the reactor system gradually decreases. As shown in Fig. 10, when the load drops, the system output power is greater than the load, which makes the rotor speed rise rapidly, and makes the compressor boost ratio and turbine expansion ratio increase. In addition, although the decrease degree of coolant temperature in the high-temperature area of the system is similar to the increase degree of coolant temperature in the low-temperature area, the density increase caused by the temperature rise of helium-xenon mixture in the low-temperature range is greater than the density decrease caused by the temperature drop of helium-xenon mixture in the high-temperature range under the same temperature change. According to the equation of gas state, the average pressure of the system shows an upward trend, so the decline rate of compressor inlet pressure is less than the rising rate of outlet pressure. Because the characteristics of compressor and turbine are different, the increase rate of boost ratio with the increase of rotation speed is greater than that of

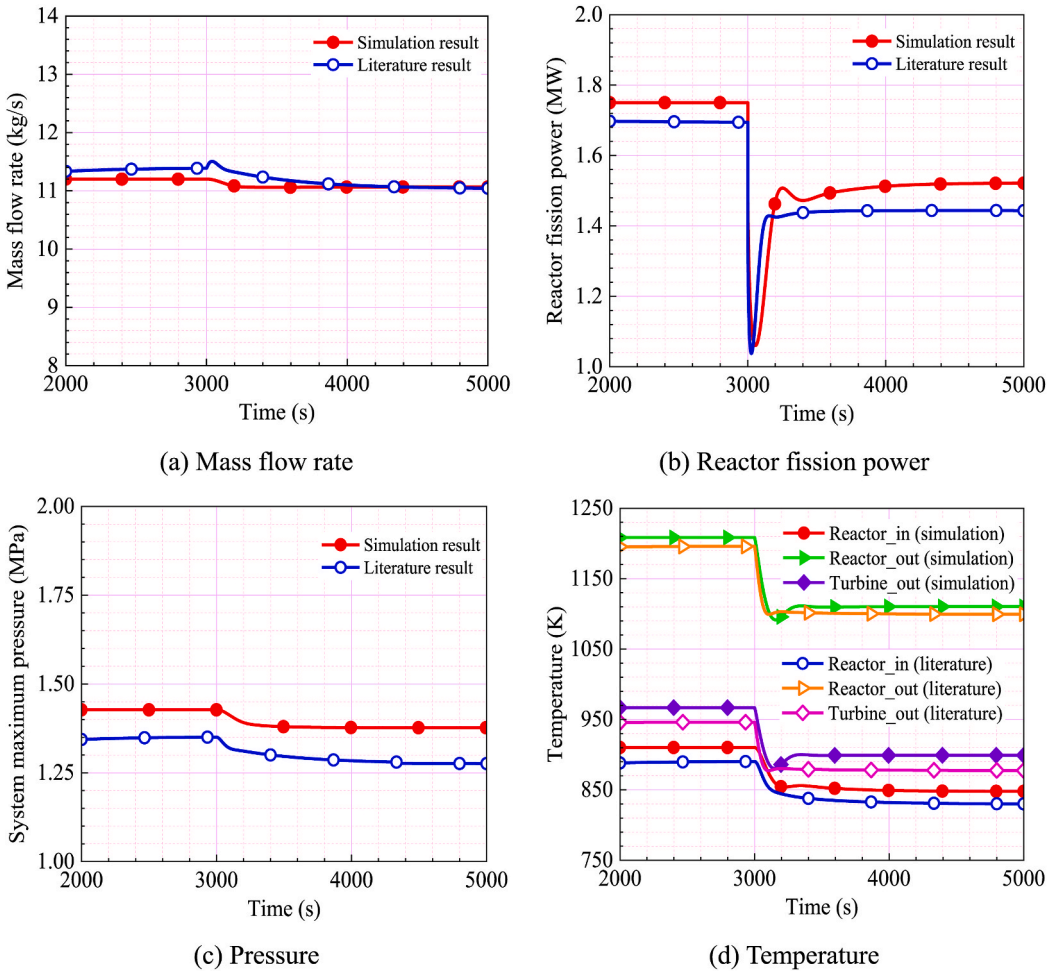


Fig. 7. Verification and comparison of transient curves under the condition of introducing negative reactivity.

expansion ratio, so the mass flow rate of compressor and turbine shows an upward trend. Increase of mass flow rate can further increase the expansion ratio and reduce the boost ratio, so as to gradually match the boost ratio and expansion ratio and make the mass flow rate stable.

$$L_{tu} = \eta_{tu} \times \int_{P_{tu,out}}^{P_{tu,in}} \frac{1}{\rho_{tu}} dP \tag{14}$$

$$W_{tu} = G_{tu} L_{tu} = -G_{tu}\left(H_{out} - H_{in} + \frac{V_{out}^2 - V_{in}^2}{2} + g(Z_{out} - Z_{in})\right) \tag{15}$$

$$L_c = \frac{\int_{P_{c,in}}^{P_{c,out}} \frac{1}{\rho_c} dP}{\eta_c} \tag{16}$$

$$W_c = G_c L_c = G_c\left(H_{out} - H_{in} + \frac{V_{out}^2 - V_{in}^2}{2} + g(Z_{out} - Z_{in})\right) \tag{17}$$

where, L_{tu} is the turbine specific power; $P_{tu,out}$ is the turbine outlet pressure; $P_{tu,in}$ is the turbine inlet pressure; η_{tu} is the turbine efficiency; ρ_{tu} is the density of helium-xenon gas mixture inside the turbine; W_{tu} is the turbine power; G_{tu} is the turbine mass flow rate; L_c is the compressor specific power; $P_{c,out}$ is the compressor outlet pressure; $P_{c,in}$ is the compressor inlet pressure; η_c is the compressor efficiency; ρ_c is the density of helium-xenon gas mixture inside the compressor; W_c is the compressor power; G_c is the compressor mass flow rate.

As shown in Fig. 11, according to Eq. (14) and Eq. (16), the specific power of turbine and compressor increases with the increase of expansion ratio and boost ratio. At the initial stage of the accident, the increase rate of boost ratio is greater than the expansion ratio, the increase rate of compressor efficiency is less than the decrease rate of turbine efficiency, and the change of compressor average density is small, while the average turbine density shows an upward trend. Considering the above factors, the rise rate of turbine specific power is lower than that of compressor. Combined with the change of system operating temperature and the change of turbine and compressor efficiency, the cycle efficiency shows a downward trend. In addition, assuming that the efficiency of turbine and compressor remains unchanged, the system temperature change caused by the change of boost ratio and expansion ratio will still reduce the cycle efficiency. Therefore, it is also concluded that when designing the rated steady-state parameters of the system, a larger expansion ratio and boost ratio are not conducive to achieving higher cycle efficiency.

As shown in Fig. 10(b) and (c), according to Eq. (15) and Eq. (17), the turbine outlet temperature decreases with the increase of specific power, and the compressor outlet temperature increases with the increase of specific power. As shown in Fig. 12, when the turbine outlet temperature decreases, first reduce the heat exchange power W_n. On the one hand, the reduction of heat exchange power W_n weakens the heating of the cold side fluid in the corresponding region and reduces the reactor inlet temperature. On the other hand, it weakens the cooling of the hot side fluid in the corresponding region and weakens the decreasing degree of $T_{h,2}$ with the decrease of turbine outlet temperature. After multi-layer weakening, the impact of the reduction of turbine outlet temperature on the cooler inlet temperature becomes small, so the change of turbine outlet temperature mainly affects the reactor inlet temperature. Similarly, the change of compressor outlet temperature mainly affects the

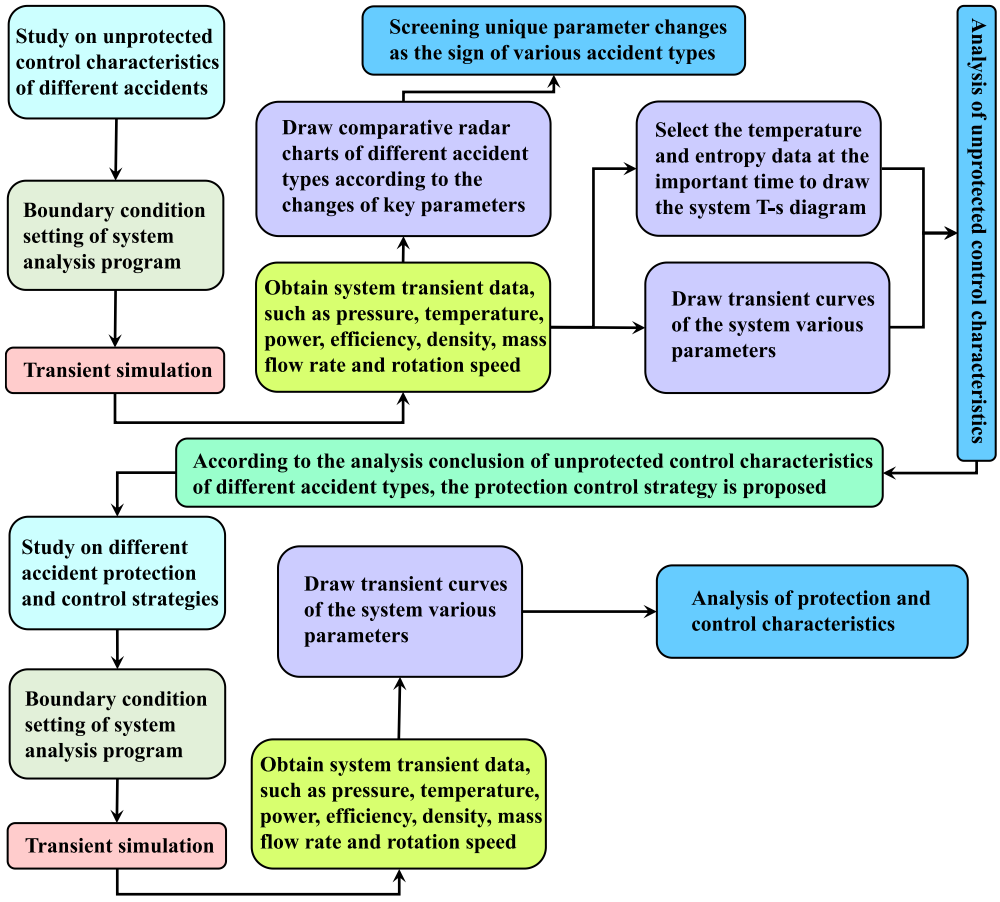


Fig. 8. Data processing flow diagram of system accident simulation.

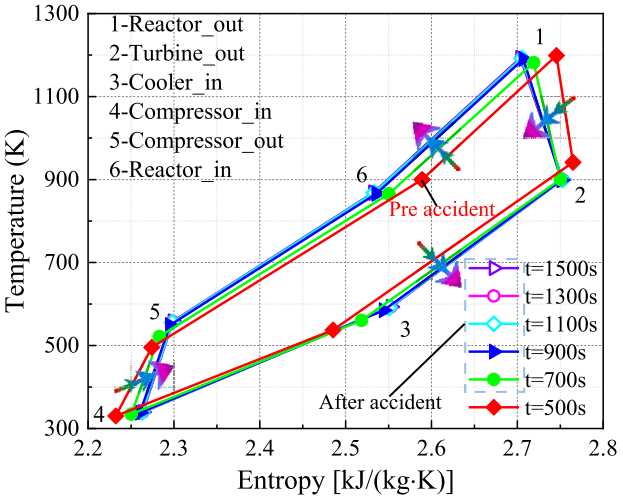


Fig. 9. Temperature-specific entropy (T-s) diagram under the unprotected condition of the 20 % step decrease in external load.

cooler inlet temperature. To sum up, the reactor inlet temperature decreases with the decrease of turbine outlet temperature, and the cooler inlet temperature increases with the increase of compressor outlet temperature. Due to the reduction of the reactor inlet temperature and the enhancement of the cooling capacity in the reactor, the heat exchange power of the reactor is increased. Therefore, the reactor outlet temperature drop is lower than the inlet temperature. The power of the cooler also increases with the increase of its inlet temperature, which weakens the rise of the compressor inlet temperature. The decreasing trend of regenerator power is caused by the increase of cold side inlet temperature and the decrease of hot side inlet temperature.

Finally, during 500s–700s, the output power of the system showed a slight upward trend due to the slow decline of cycle efficiency and the rapid rise of reactor power. After 700s, because the cycle efficiency decreased rapidly and the reactor power increased slowly, the output power showed a significant downward trend, and finally matched with the external load to stabilize the rotation speed.

It can be seen from the foregoing that the step drop of external load will lead to the rapid rise of rotation speed, system maximum pressure and reactor power. As shown in Figs. 13 and 14, when the step decrease degree of external load increases, the rise rate of rotation speed, system maximum pressure and reactor power also increases. At 630s, the rotation speed, system maximum pressure and reactor power at 100 % load loss condition increased by 17.1 %, 18.5 % and 11.1 % respectively compared with 20 % load loss condition. Therefore, the greater the degree of load loss, the greater the threat to the safety of the reactor system. The characteristic curve of compressor and turbine in this paper is established with reference to Ref. [13]. The maximum rotation speed of the characteristic curve of compressor and turbine in Ref. [13] is 125 % of the rated design rotation speed. The rated design rotation rate in this paper is 20000 rpm, so 25000 rpm is taken as the overspeed limit. When the step drop of external load exceeds 60 %, the rotation speed will exceed the limit range, which may pose a safety threat to the reactor system.

From the transient simulation results of reactor system without protection control under loss of external load accident, it can be seen that excessive reduction of external load will lead to rotor overspeed, so it is necessary to take corresponding safety protection and control

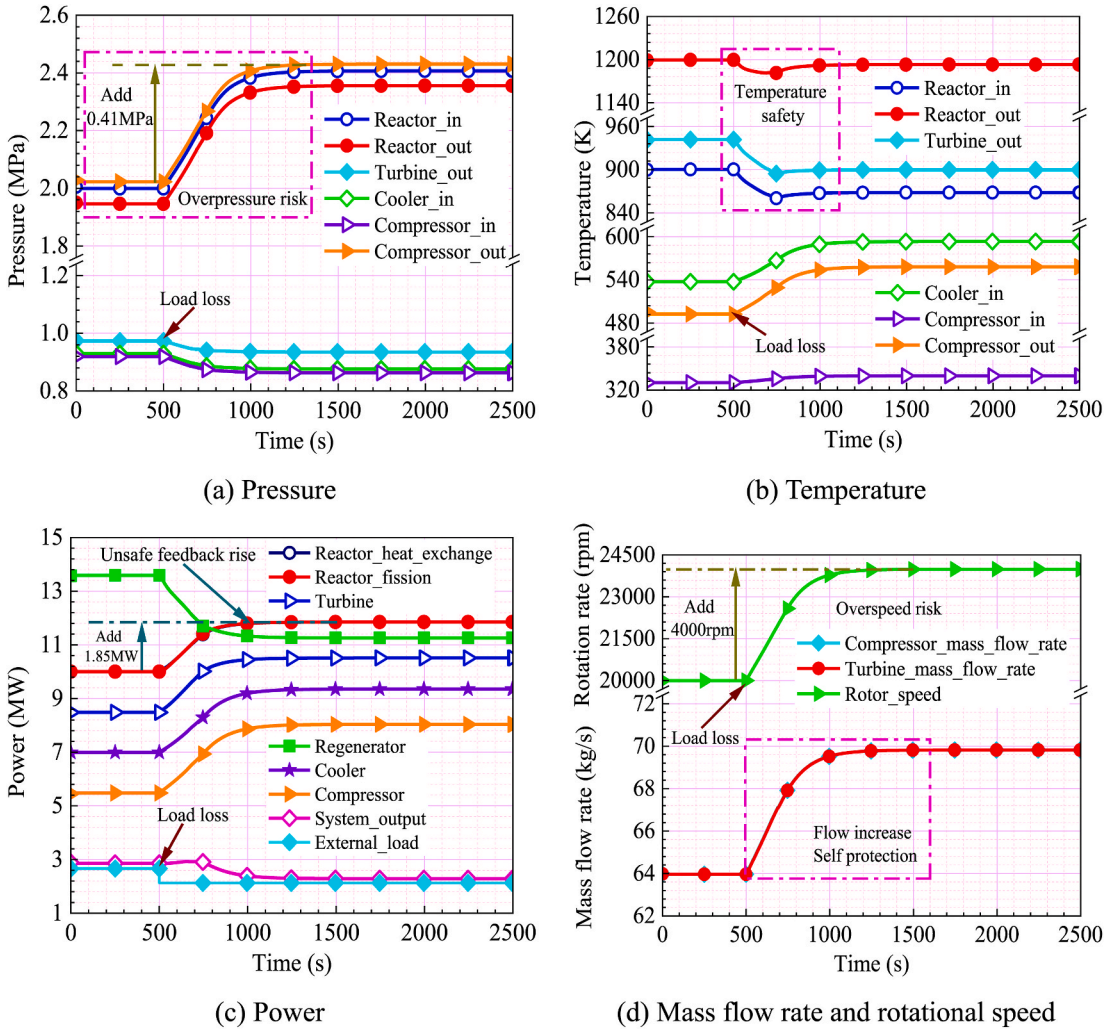


Fig. 10. Transient curve of system operating parameters under the unprotected condition of the 20 % step decrease in external load.

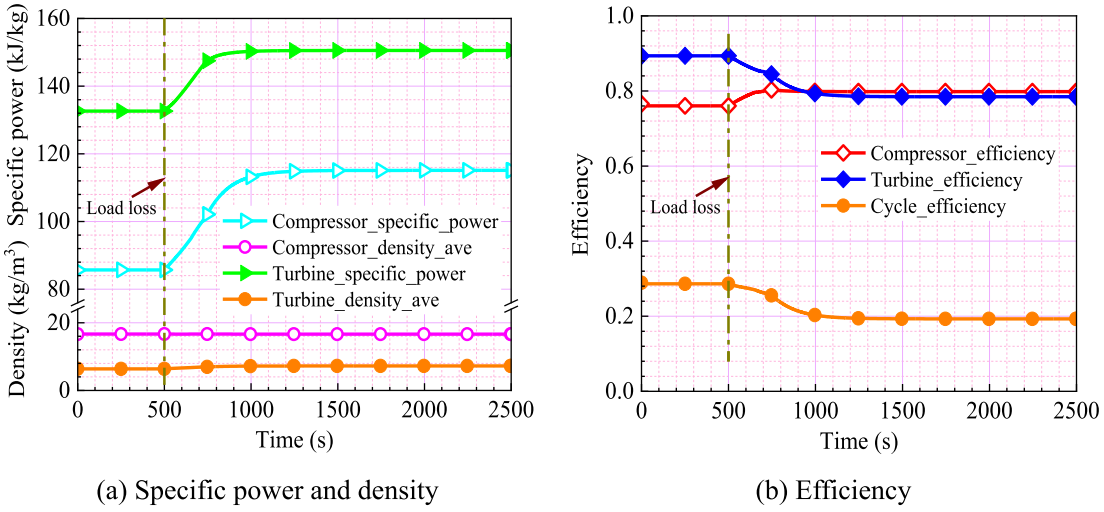


Fig. 11. Transient curve of rotating mechanical parameters under the unprotected condition of the 20 % step decrease in external load.

measures. For the control measures to restrain the speed rise rate, the following methods can be adopted mainly from increasing the compressor power and reducing the turbine power: introducing negative reactivity, bypass control and reducing the cooling water flow. However, reducing the cooling water flow will reduce the waste heat discharge capacity of the system, which may further expand the accident hazards, so this method is unreasonable. Under various load loss degrees, the accident with 100 % load loss is the most harmful, so this

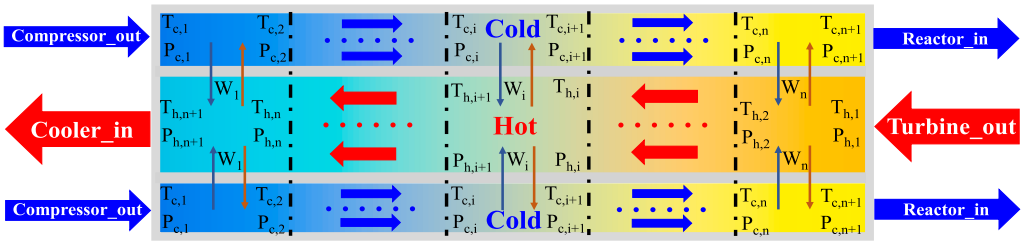


Fig. 12. Regenerator schematic diagram.

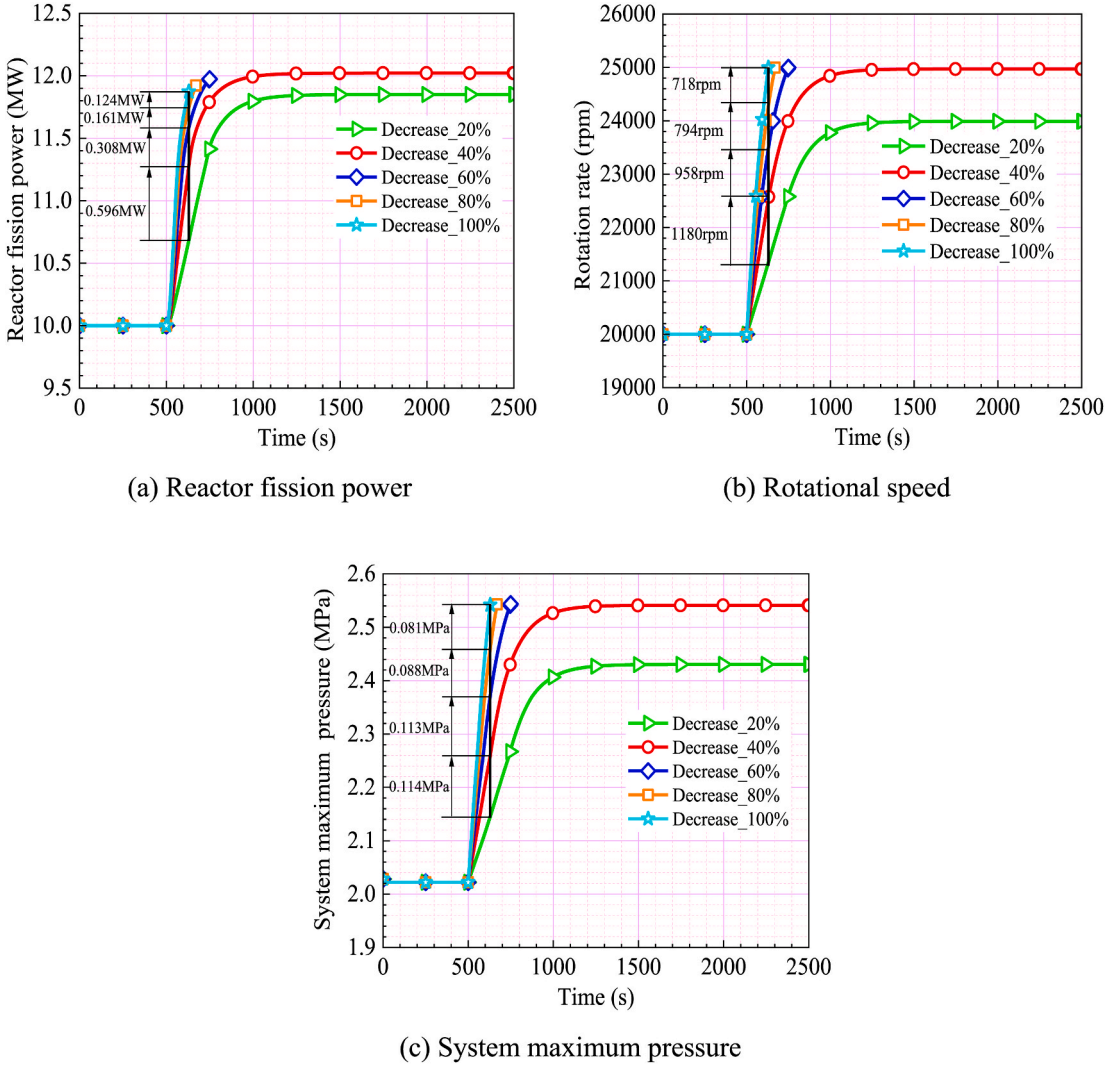


Fig. 13. Transient curves of key system safety parameters under different external load loss unprotected conditions.

section takes the case of 100 % load loss as an example to study the characteristics of introducing negative reactivity and bypass control.

For the introduction of negative reactivity control strategy, the initial state of the system simulation is the rated steady state condition, and the load power is reduced by 100 % at 500s. When the overspeed signal is triggered by excessive rotating speed, a large negative reactivity is introduced into the reactor. In addition, if the maximum pressure of the system exceeds the limit value, the volume discharge valve will execute the command of Table 1. The specific protection action response sequence is shown in Table 2, and the simulation results are shown in Fig. 15.

As shown in Fig. 15, when the negative reactivity was introduced at 600s, the fission power of the reactor dropped sharply, resulting in a rapid decline in the heat exchange power of the reactor and a rapid reduction in the outlet temperature of the reactor. The output power of the system is the difference between the turbine power, the compressor power and the friction loss power. As the reactor outlet temperature decreases, the turbine power also decreases rapidly, and is smaller than the compressor for a certain time, making the system output power negative. When the output power of the system is less than 0, the turbine power cannot fully compensate the power consumption of the compressor. At this time, part of the compressor power consumption is provided by the conversion of rotor kinetic energy, which makes the rotation speed drop rapidly. However, the decrease of rotation speed

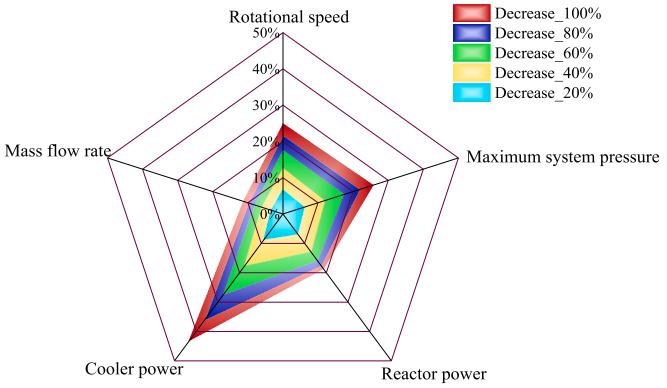


Fig. 14. Percentage change of main parameters at 630s under different load loss degrees.

Table 2
Protection action response sequence under the introducing negative reactivity.

Time	Action
0~500s	Rated steady-state operation of the system
500s	Loss of all external load
600s	Trigger overspeed signal and introduce −130pcm reactivity

makes the compressor power drop rapidly, and the output power of the system changes from a downward trend to an upward trend, and finally stabilizes at 0. At this time, the compressor power and turbine power are roughly equal, and the rotation speed is stable at about 21500 rpm. The compressor boost ratio and turbine expansion ratio are proportional to the rotation speed, but the change rate of the boost ratio with the rotation speed is greater than the expansion ratio, so that the mass flow rate and the maximum pressure of the system have the same trend with the rotation speed, and the minimum pressure of the system has the opposite trend with the rotation speed. The peak value of the maximum pressure of the system did not trigger the overpressure signal, and the volume discharge protection control was not put into operation.

As shown in Fig. 16, the bypass control strategy is realized through the bypass valve and its pipeline. When the bypass valve is opened, part

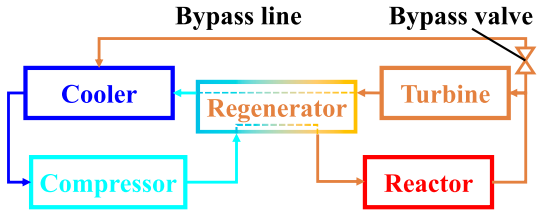
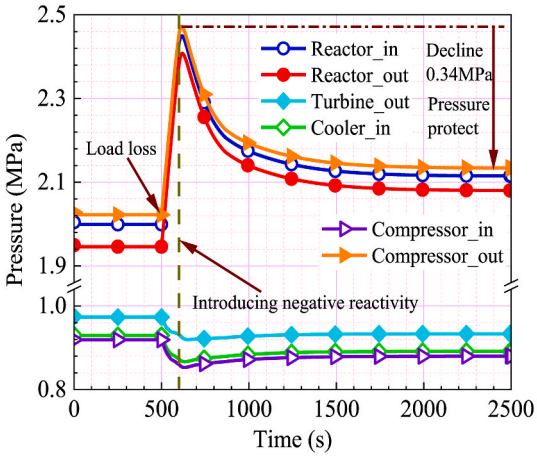
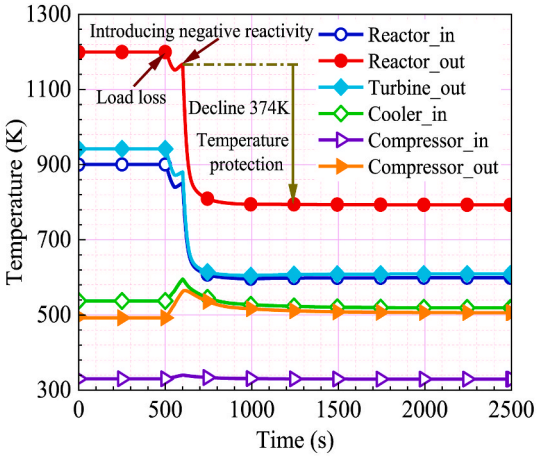


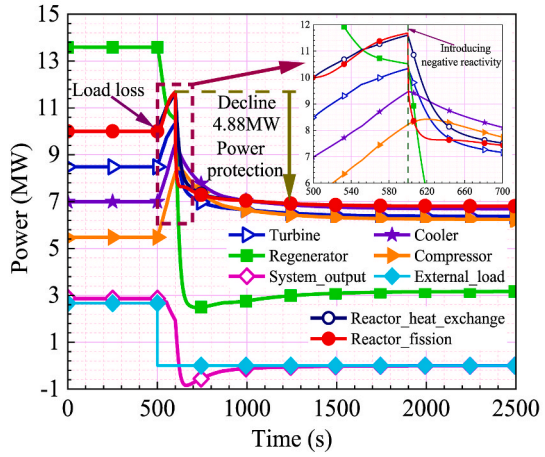
Fig. 16. Schematic diagram of bypass control principle.



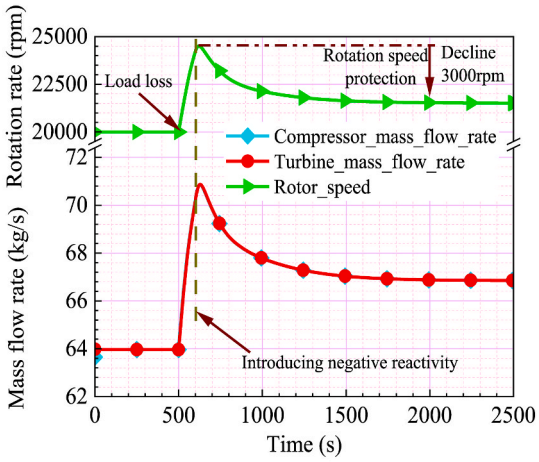
(a) Pressure



(b) Temperature



(c) Power



(d) Mass flow rate and rotational speed

Fig. 15. Transient curve of system operating parameters under the protected condition of the introducing negative reactivity.

of the coolant flows directly into the cooler from the reactor outlet, so as to reduce the turbine mass flow rate and achieve the purpose of reducing the turbine power and suppressing the speed rise. The initial state of the system simulation is the rated steady-state condition. The load power is reduced by 100 % at 500s. When the overspeed signal is triggered by excessive speed, the opening of the bypass valve is linearly increased to 18 % within 10 s. In addition, if the maximum pressure of the system exceeds the limit value, the volume discharge valve will execute the command of Table 1. The specific protection action response sequence is shown in Table 3, and the simulation results are shown in Fig. 17.

As shown in Fig. 17, when the bypass valve is opened within 600s–610s, the bypass coolant mass flow rate increases rapidly, and then decreases with the pressure difference between the reactor outlet and the cooler inlet. Due to the sudden increase of bypass mass flow rate, the mass flow rate of compressor and turbine increases and decreases respectively, and then decreases together with the decrease of rotation speed. The compressor power increases rapidly with the sudden increase of mass flow rate. Similarly, the turbine power decreases rapidly, which makes the system output power drop rapidly below 0, thus reducing the rotation speed. As the speed decreases, the compressor power decreases rapidly, which makes the system output power rise. Finally, the compressor and turbine power cancel each other and maintain the speed at about 20800 rpm.

After the bypass valve is opened, due to the rapid increase of mass flow rate in the reactor, the reactor outlet temperature shows a rapid downward trend. The turbine outlet temperature and reactor inlet temperature both show a rapid downward trend in the same way as the regenerator described above. The high temperature coolant at the reactor outlet flows into the cooler inlet through the bypass pipe, making the cooler inlet temperature rise rapidly, and the compressor inlet and outlet temperatures rise accordingly. Subsequently, due to the decrease of rotation speed, the turbine specific power and compressor specific power decreased, the high temperature area of the system changed from a downward trend to an upward trend, and the low temperature area of the system changed from an upward trend to a downward trend. The compressor boost ratio is inversely correlated with the mass flow rate, and the turbine expansion ratio is positively correlated with the mass flow rate, and the rate of expansion ratio changing with the mass flow rate is greater than the boost ratio. Therefore, when the mass flow rate of compressor and turbine rises and falls sharply respectively, the decrease of expansion ratio is larger, and the rise rate of maximum system pressure (compressor outlet pressure) increases. At 603s, the maximum pressure of the system exceeds 2.5 MPa, and the volume discharge valve is opened by 70 %, which makes the pressure drop rapidly and realizes pressure protection. At 620s, the maximum pressure of the system is lower than 2.1 MPa, the volume discharge valve is closed, and the maximum pressure of the system decreases only due to the decrease of the rotation speed, and its decline rate decreases.

To sum up, the introduction of negative reactivity and bypass control can both inhibit the rapid rise of rotation speed and maintain the rotation speed at another stable value, so as to achieve the stability of system parameters and accident protection. After the bypass control protection strategy is implemented, the maximum pressure of the system will suddenly increase, resulting in pressure overrun, and the reactor power will be at a high value after stabilization. However, after the introduction of reactivity protection strategy, the maximum pressure of the system immediately decreased without the risk of pressure overrun. After the reactor power and outlet temperature stabilized, they were at a low value, which were 55.46 % and 66.69 % of the bypass control protection strategy, respectively. The comparison of the two protection control performances is shown in Fig. 18. Therefore, from the perspective of thermal safety and economy, the introduction of reactivity is better than bypass control, but this study only considers from the thermal aspect, and does not consider whether the introduction of reactivity will cause reactor power oscillation or other harm. Therefore, the advantages and disadvantages of the two protection strategies need to be further studied.

Table 3
Protection action response sequence under the bypass control.

Time	Action
0~500s	Rated steady-state operation of the system
500s	Loss of all external load
600s	Trigger overspeed signal and linearly increase the opening of bypass valve to 18 % within 10s
603s	Trigger pressure exceeding signal (>2.5 MPa) and step increase in volume discharge valve opening to 70 %
620s	Trigger the pressure relief end signal (<2.1Mpa) and close the volume discharge valve

4.2. Analysis of unprotected transient characteristics and protection control strategy characteristics under heat sink partial loss accident

The partial loss of heat sink accident is one of the typical accidents of nuclear power plant. During the rated steady-state operation of the system, the mass flow rate of cooling water at the secondary side of the cooler may drop rapidly due to the loss of partial driving force, resulting in the partial loss of heat sink accident. Therefore, in this section, the transient response process of the reactor Brayton system after the step drop of the cooling water driving pressure at the secondary side of the cooler by 43.5 % is simulated, and the transient characteristics of the partial loss of heat sink accident of the system are analyzed. The initial state of the system simulation is the rated steady-state condition. At 500s, the driving pressure of the cooling water drops 43.5 %, and the external load remains constant. The simulation results are shown in Figs. 20 and 21, and the temperature-specific entropy (T-s) diagrams is shown in Fig. 19.

As shown in Figs. 20 and 21, when the driving pressure of cooling water drops, the mass flow rate of cooling water drops rapidly from 60 kg/s to 33 kg/s, which makes the power of cooler drop rapidly and the inlet and outlet temperatures of compressor rise rapidly. In the same way as the recuperator described in 4.1, the inlet temperature of the cooler also rises rapidly, making the cooler power change from a downward trend to an upward trend. When the inlet and outlet temperature of compressor and the inlet temperature of cooler rise, the system pressure rises due to the expansion of coolant. Because the system pressure increases, the pressure difference between the inlet and outlet of the compressor and the turbine also increases when the boost ratio and the turbine change little. However, because the boost ratio is greater than the expansion ratio, the pressure difference between the inlet and outlet of the compressor increases more than that of the turbine. Combined with the influence of compressor and turbine efficiency, it can be seen from Eq. (14) and Eq. (16) that the specific power of the compressor and the turbine step increases at 500s, and the step increases of the specific power of the compressor is greater. At the moment of 500s, the mass flow rate of compressor and turbine changed little, so the output power decreased, and the rotation speed began to show a downward trend.

As the rotation speed continues to decline, the turbine expansion ratio and compressor boost ratio continue to decline, making their specific power continue to decline. Therefore, the turbine outlet temperature shows an upward trend, the compressor outlet temperature shows a downward trend, and the other temperatures show a corresponding change trend due to the role of the regenerator. The maximum pressure and minimum pressure of the system show a downward trend and an upward trend respectively due to the continuous decline of expansion ratio and boost ratio. The system mass flow rate has the characteristic of decreasing with decreasing rotation speed. With the continuous increase of the reactor inlet temperature, the cooling and heat transfer capacity in the reactor deteriorates, the reactor heat exchange power continues to decline, and the reactor fission power also decreases with the heat exchange power due to temperature feedback.

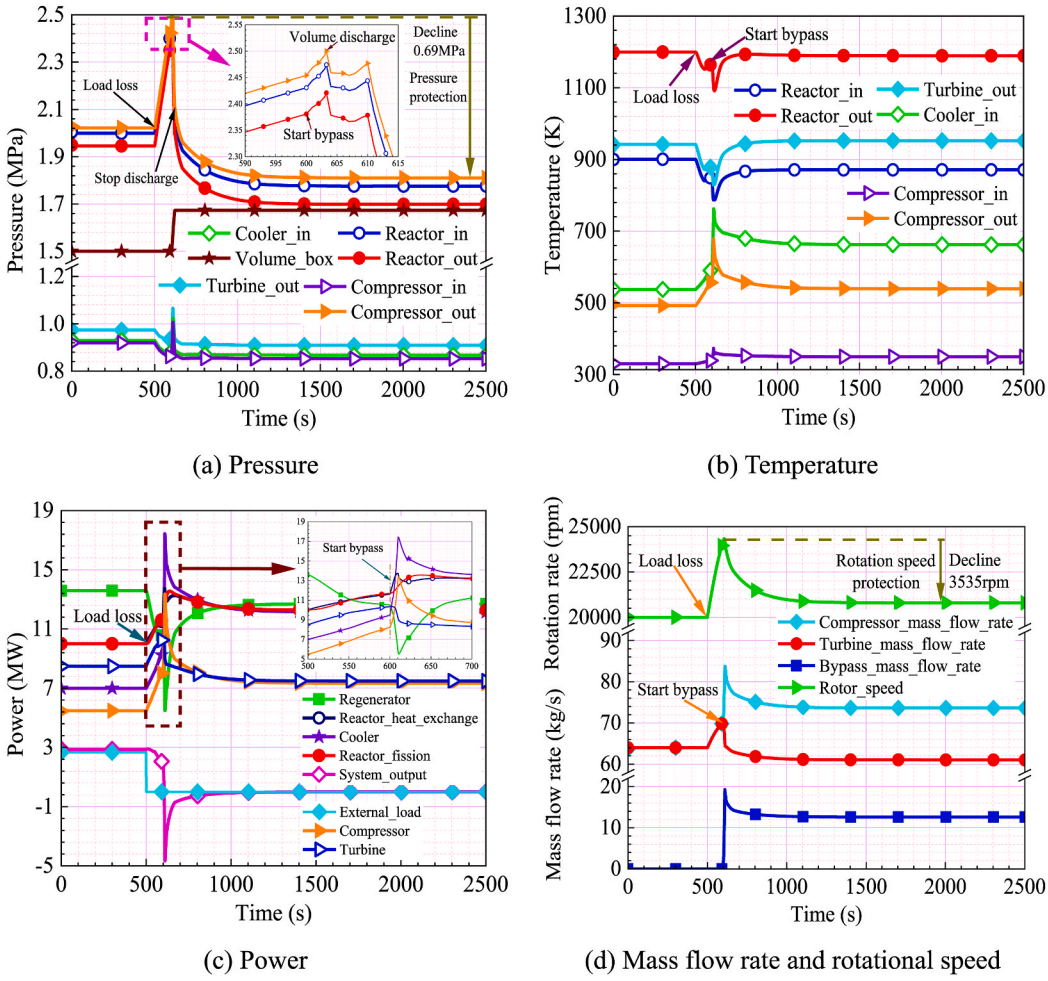


Fig. 17. Transient curve of system operating parameters under the protected condition of the bypass control.

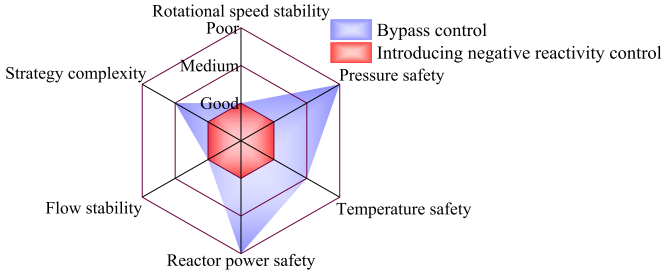


Fig. 18. Performance comparison diagram of introducing negative reactivity and bypass control.

Although the circulation efficiency showed a slight upward trend in 500s–900s, the system output power still showed a downward trend due to the continuous decline of reactor power. When the rotation speed and mass flow rate deviate greatly from the design value, the compressor efficiency begins to drop sharply, which makes the cycle efficiency drop sharply, and accelerate the decline of system output power, rotation speed and mass flow rate. Due to the decrease of system mass flow rate and the gradual loss of turbine cooling capacity, the temperature in the reactor rises rapidly, with the risk of temperature exceeding the limit, and corresponding protective measures need to be taken to control the temperature rise rate in the reactor.

From the transient simulation results of reactor system without protection control under heat sink partial loss accident, it can be seen that after the cooling water mass flow decreases, the rotation speed will

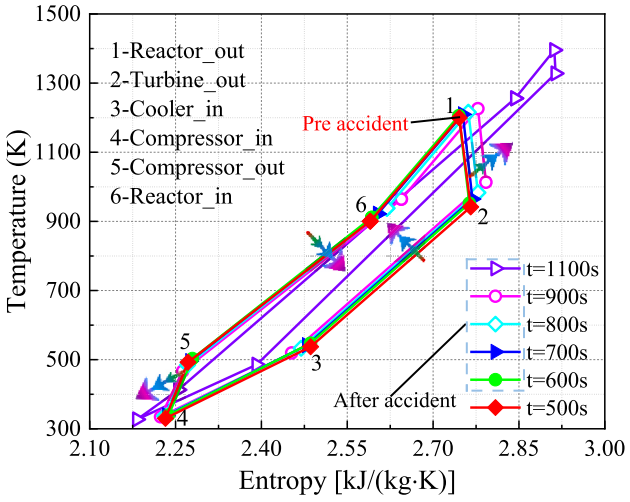


Fig. 19. Temperature-specific entropy (T-s) diagram under the unprotected condition of the partial loss of heat sink.

continue to decline, the cooling capacity of the turbine will gradually weaken, and the reactor will have the risk of temperature overrun due to the continuous heat recovery through the regenerator. The decrease of rotation speed will reduce the cycle driving force and cause the decrease of circulating mass flow rate. Although the change of the system

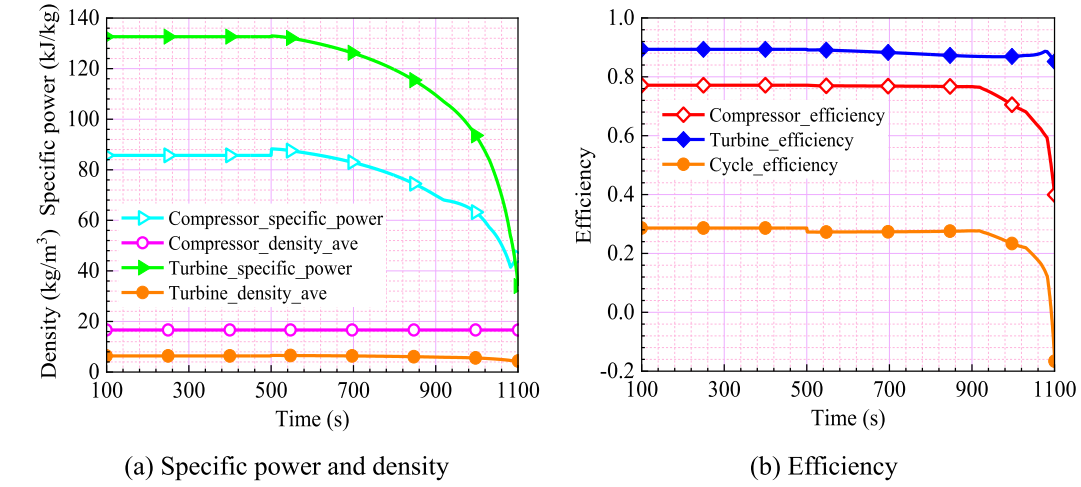


Fig. 20. Transient curve of rotating mechanical parameters under the unprotected condition of the partial loss of heat sink.

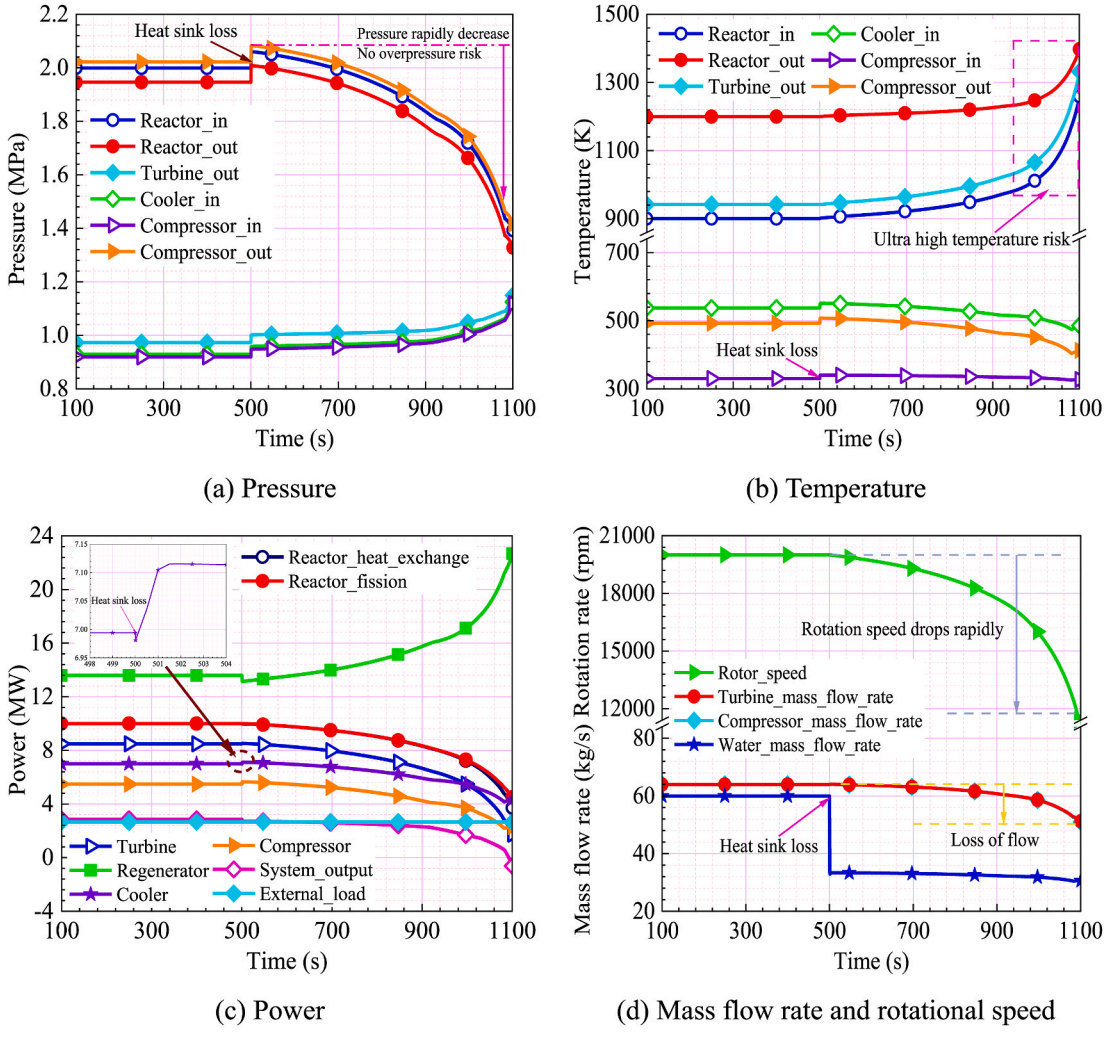


Fig. 21. Transient curve of system operating parameters under the unprotected condition of the partial loss of heat sink.

temperature and pressure causes the change of the flow resistance in the circulating circuit, as long as the rotation speed is stabilized, the circulating mass flow rate can be stabilized automatically according to the characteristics of the compressor and turbine. The characteristics are that the boost ratio of compressor increases with the decrease of mass flow rate and the expansion ratio of turbine decreases with the decrease of mass flow rate. In addition, the circulating flow resistance will also decrease with the decrease of mass flow rate. Therefore, the key problem to alleviate the accident is to suppress the rate of decline of rotation speed and make it stable. The load following the change of output power

of the system can effectively inhibit the decline of rotation speed. This section will study the characteristics of the load following protection control strategy. The accident introduction process is consistent with unprotected control. The load in the whole simulation process has the ability to follow the system output power. In addition, if the maximum pressure of the system exceeds the limit value, the volume discharge valve will execute the command in Table 1. The simulation results are shown in Fig. 22.

As shown in Fig. 22, in the load following state, when the output power of the system step drops, the load also decreases, which inhibits the decline of rotation speed. The external load can match the system output power in a short time to maintain the rotation speed at another stable value. When the rotation speed is stable, the system mass flow rate and the other parameters will also be stabilized quickly, and the system pressure will increase only due to the increase of temperature. Due to the low pressure rise value, the maximum pressure of the system did not trigger the overpressure signal, and the volume discharge valve was not put into operation.

Similarly to unprotected control, the inlet and outlet temperatures of compressor and the inlet temperature of cooler increase rapidly due to the step decrease of cooling water mass flow rate. The reactor inlet and outlet temperature and turbine outlet temperature first increased due to the decrease of rotation speed, then decreased due to the stability of rotation speed and the decrease of reactor power, and then stabilized quickly. Through the load following protection strategy, after the partial

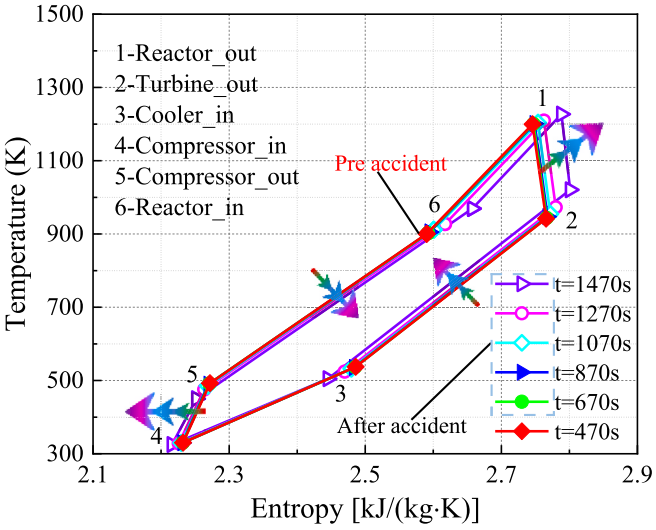


Fig. 23. Temperature-specific entropy (T-s) diagram under the unprotected condition of the opening of the main valve is reduced by 20 %.

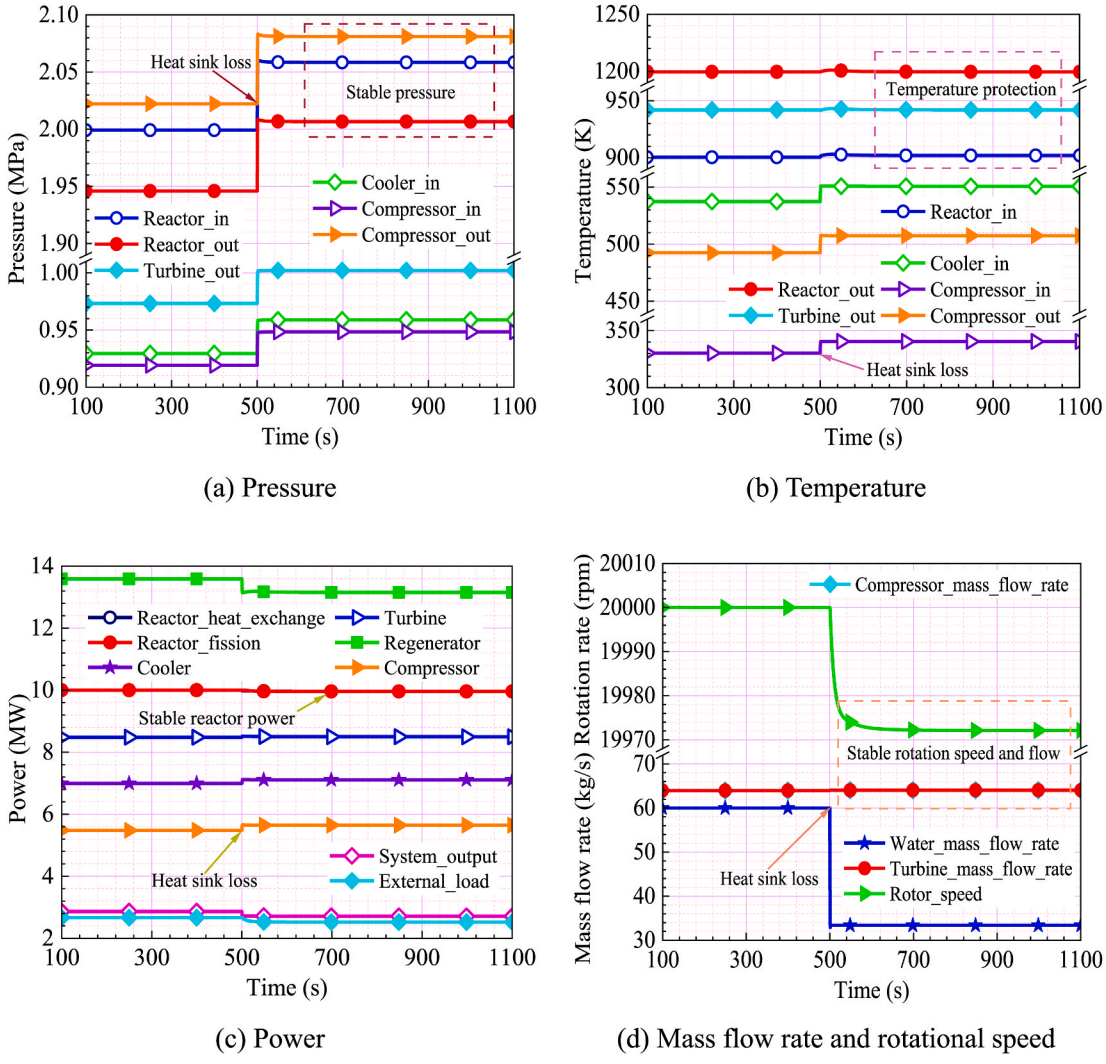


Fig. 22. Transient curve of system operating parameters under the protected condition of the load following protection control.

loss of heat sink accident, the system parameters will soon stabilize after a small fluctuation, and the overall change is small. Therefore, when a partial loss of heat sink accident occurs, the system can be stable in another steady state in a very short time under the load following protection state, achieving temperature protection, and the accident risk is greatly weakened. This also proves that the load following protection strategy can achieve the safety protection of the reactor system under the partial loss of heat sink accident.

4.3. Analysis of unprotected transient characteristics and protection control strategy characteristics under main valve accidental closure accident

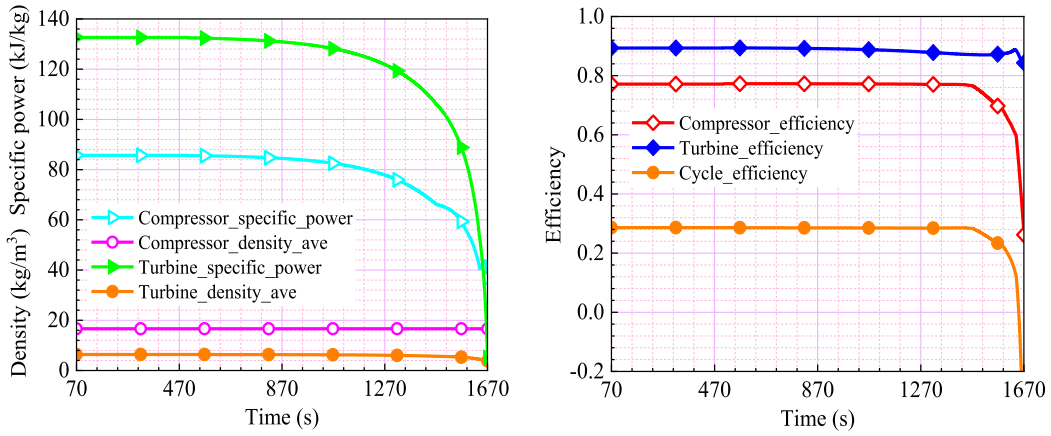
During the rated steady-state operation of the system, the valve opening of the main pipeline in front of the reactor inlet may be reduced due to valve failure, which increases the coolant circulation resistance, resulting in the main valve accidental closure accident. Therefore, this section takes the main valve opening decreasing by 20 % as an example to simulate the transient response process of the reactor Brayton system and analyze the transient characteristics of the system main valve accidental closure accident. The initial state of the system simulation is the rated steady-state condition. Within 500s–510s, the opening of the main valve is reduced by 20 %, and the external load remains constant. The simulation results are shown in Figs. 24 and 25, and the temperature-specific entropy (T-s) diagrams is shown in Fig. 23.

As shown in Figs. 24 and 25, when the opening of the main valve is gradually reduced by 20 %, the pressure drop of the main valve increases and the inlet and outlet pressure of the reactor is reduced. Therefore, the expansion capacity of the turbine is weakened and the power of the turbine is reduced, which makes the output power of the system decrease slightly and the rotation speed begins to decrease. Before 870s, due to the slow decline of turbine mass flow rate, specific power and efficiency, the system output power decreased slowly and the rotation speed decreased slightly. After 870s, the decline rate of the system output power increased with the increase of the decline rate of the reactor power, and the rotation speed began to decrease rapidly. When the rotation speed and mass flow rate differ greatly from the design value, the compressor efficiency drops sharply, which makes the cycle efficiency drop sharply, the system output power also drops sharply, and the rotation speed drop rate further increases.

The change trend and principle of other parameters of the system due to rotation speed drop are consistent with 4.2. This section will not repeat the description. Therefore, like the partial loss of heat sink accident, the main valve accidental closure accident also has the risk of reactor temperature exceeding the limit due to the decrease in rotation speed and loss of mass flow rate, and corresponding protective measures need to be taken to control the temperature rise rate in the reactor. As shown in Figs. 26 and 27, it can be seen from the above that the main valve be closed incorrectly, which will lead to the continuous decrease of the rotation speed. When the decrease percentage of main valve opening increases, the decrease rate of system mass flow rate and rotation speed and the rise rate of reactor outlet temperature also increase. At 1160s, the rotation speed and mass flow rate at the 40 % misclosure condition decreased by 59.3 % and 30.4 % respectively compared with the 10 % misclosure condition, reactor outlet temperature increased by 28.9 %. Besides, the rotational speed, mass flow rate, and reactor outlet temperature reached their limit values about 740 s earlier. Therefore, a larger false closing degree of the main valve will worsen and accelerate the development of the accident, so that the reactor can reach the limit value in a shorter time, and the accident hazard is greater.

From the transient simulation results of reactor system without protection control under main valve accidental closure accident, it can be seen that the rotation speed will continue to drop after the main valve is closed by mistake. In the same way as 4.2, the key to mitigate the accident is to suppress and stabilize the rotation speed drop rate. The load following the change of the output power of the system can effectively inhibit the decline of the rotation speed. In the main valve accidental closure accident, a small reduction in the opening of the main valve is common. In this section, a 20 % reduction in the opening of the main valve is comprehensively selected as a case study of the characteristics of the protection control strategy to study the characteristics of the load following protection control strategy. The accident introduction process is consistent with unprotected control. The load in the whole simulation process has the ability to follow the system output power. In addition, if the maximum pressure of the system exceeds the limit value, the volume discharge valve will execute the command in Table 1. The simulation results are shown in Fig. 28.

As shown in Fig. 28, in the load following state, when the output power of the system decreases, the load also decreases, which inhibits the decline of rotation speed. The external load can match the system output power in a short time to maintain the rotation speed at another stable value. When the rotation speed is stable, the parameters of the system are also stable quickly, and the pressure at the inlet and outlet of the reactor decreases due to the increase of the resistance pressure drop of the main valve. The maximum pressure of the system did not trigger the overpressure signal, and the volume discharge valve was not put into operation. The circulating flow resistance introduced by the main valve accidental closure makes the circulating mass flow rate drop slightly. However, since the rotation speed is stabilized, the circulating mass flow rate can be stabilized quickly according to the characteristics of the



(a) Specific power and density (b) Efficiency

Fig. 24. Transient curve of rotating mechanical parameters under the unprotected condition of the opening of the main valve is reduced by 20 %.

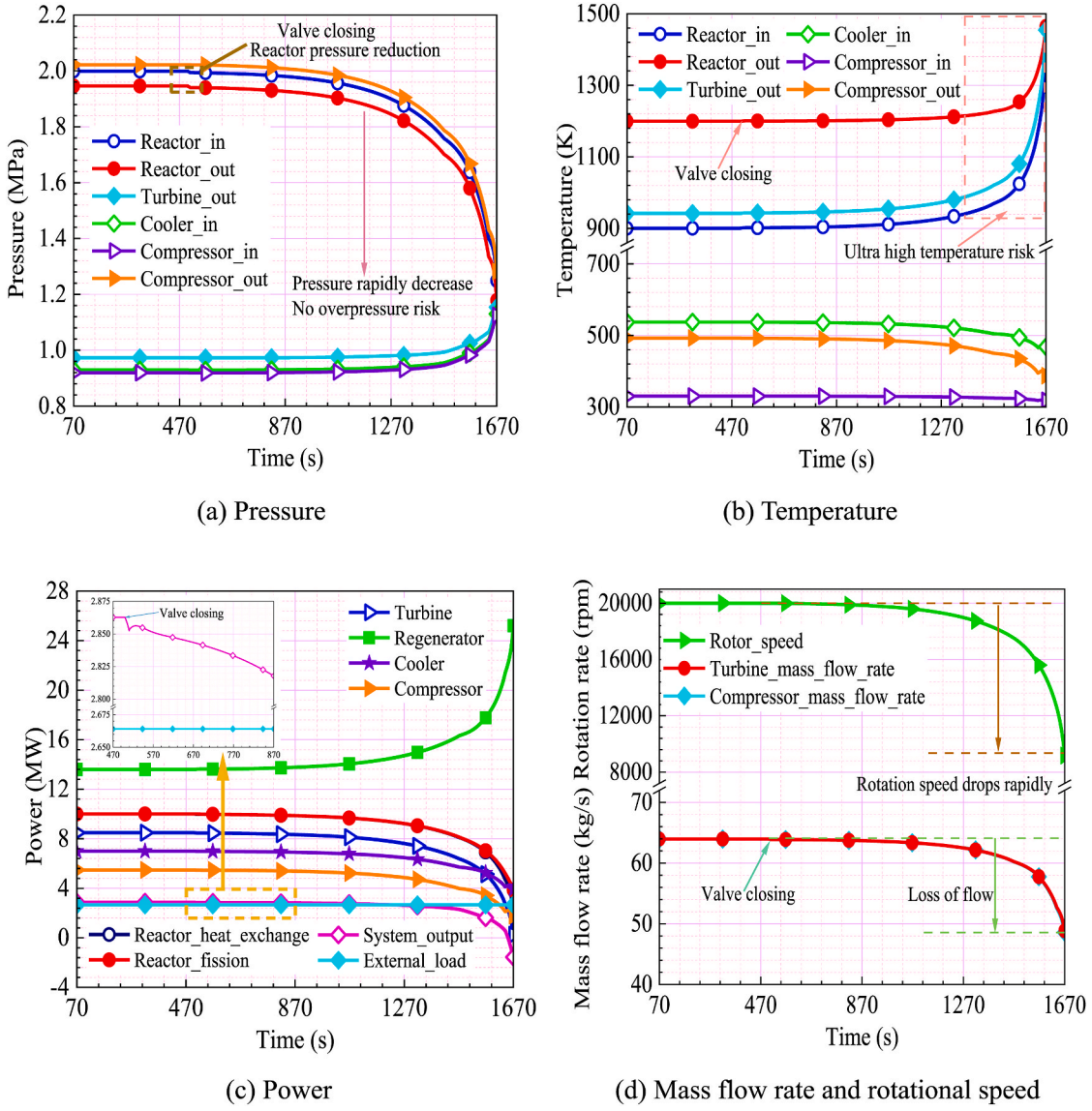


Fig. 25. Transient curve of system operating parameters under the unprotected condition of the opening of the main valve is reduced by 20 %.

compressor and turbine.

Similarly to 4.2, the reactor inlet and outlet temperatures and turbine outlet temperatures showed a trend of first increasing and then decreasing, and then stabilized soon. Through the load following protection strategy, after the main valve accidental closure accident, the system parameters will soon stabilize after a small fluctuation, and the overall change is small. Realize temperature protection and avoid the risk of ultra-high temperature. Therefore, when the main valve is closed by mistake, the system can be stable in another steady state in a very short time under the load following protection state, and the accident risk is greatly weakened. This also proves that the load following protection strategy can realize the safety protection of the reactor system in the main valve accidental closure accident.

4.4. Analysis of unprotected transient characteristics and protection control strategy characteristics under brayton unit efficiency step decrease accident

During the rated steady-state operation of the system, the work loss of the compressor and turbine may increase due to the internal component failure, and the efficiency of the compressor and turbine may be step reduced, resulting in the Brayton unit efficiency step decline accident. Therefore, in this section, the transient response process of the reactor Brayton system after the compressor and turbine efficiency step drop of 0.05 is simulated and the transient characteristics of the system Brayton unit efficiency step drop accident are analyzed. The initial state of the system simulation is the rated steady-state condition. At 500s, the compressor and turbine efficiency step decreases by 0.05, and the external load remains constant. The simulation results are shown in Figs. 30 and 31, and the temperature-specific entropy (T-s) diagrams is shown in Fig. 29.

As shown in Figs. 30 and 31, when the efficiency of compressor and turbine step decreases by 0.05, the specific power of compressor and turbine increases and decreases respectively. It makes the system output power step down and the turbine outlet temperature and compressor outlet temperature step up. When the output power of the system step drops, it cannot match with the external load, making the rotation speed drop rapidly. The system pressure step increases with the increase of temperature, and then changes with the change of rotation speed.

The change trend and principle of other parameters of the system due to rotation speed drop are consistent with 4.2. This section will not repeat the description. Therefore, like the partial loss of heat sink accident, the Brayton unit efficiency step decline accident also has the risk of reactor temperature exceeding the limit, and corresponding protective

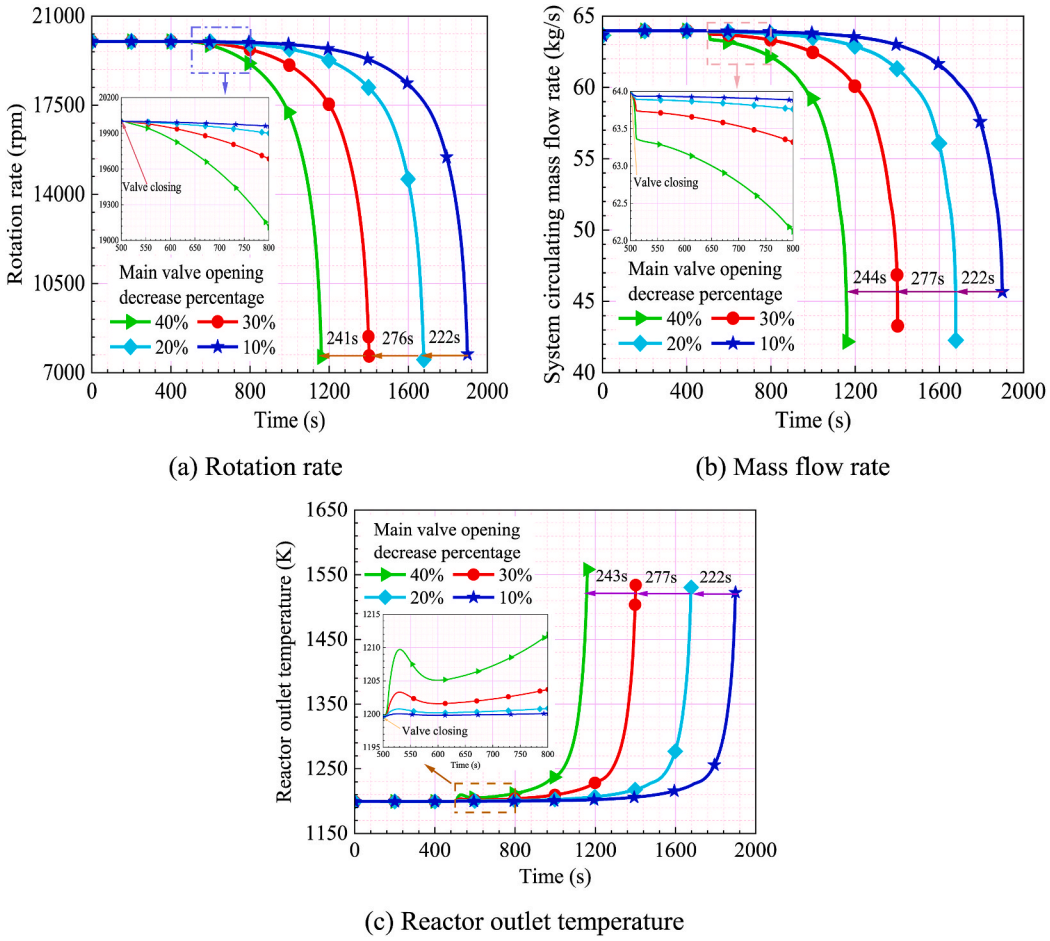


Fig. 26. Transient curves of key system parameters under different main valve opening reduction value conditions.

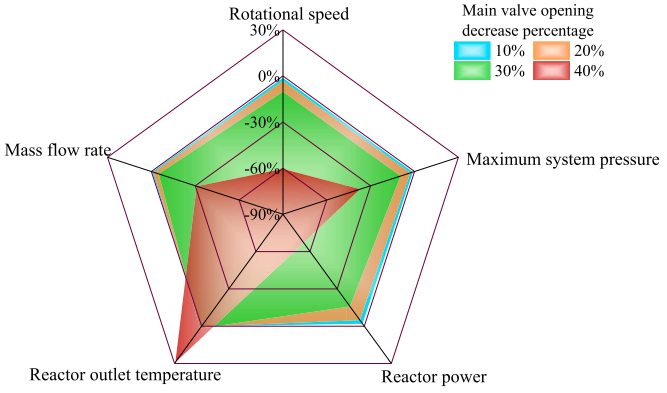


Fig. 27. Percentage change of main parameters at 1160s under different main valve opening decrease degrees.

measures need to be taken to control the temperature rise rate in the reactor.

From the transient simulation results of reactor system without protection control under Brayton unit efficiency step decrease accident, it can be seen that the step reduction of the efficiency of the Brayton unit will lead to the continuous decline of the rotation rotor. In the same way as 4.2, the key to mitigate the accident is to suppress and stabilize the rotation speed drop rate. The load following the change of the output power of the system can effectively inhibit the decline of the rotation speed. This section will study the characteristics of the load following protection control strategy. The accident introduction process is consistent with unprotected control operating condition. The load in the whole simulation process has the ability to follow the system output power. In addition, if the maximum pressure of the system exceeds the limit value, the volume discharge valve will execute the command in Table 1. The simulation results are shown in Fig. 32.

As shown in Fig. 32, in the load following state, when the output power of the system step drops, the load also decreases, which inhibits the decline of rotation speed. After a period of time, the matching of external load and system output power maintains the rotation speed at another stable value. When the rotation speed is stable, the parameters of the system will also be stable soon. It can be seen from unprotected control that the temperature at each part of the system step increases due to the step decrease of turbine specific power and the step increase of compressor specific power, and the system pressure increases due to the expansion of temperature. Due to the low pressure rise value, the maximum pressure of the system did not trigger the overpressure signal, and the volume discharge valve was not put into operation. As the rotation speed drop is small, the change is inhibited and tends to be stable, and the system mass flow rate also tends to be stable after a small reduction.

The inlet and outlet temperatures of compressor and the inlet temperature of cooler showed a downward trend after a step increase due to the decrease of rotation speed, and finally stabilized with the stability of rotation speed. The reactor inlet and outlet temperature and turbine outlet temperature increased first and then decreased due to the decrease of rotation speed and reactor power, and finally stabilized with the stability of rotation speed and reactor power. Through the load following protection strategy, after the efficiency step down accident of Brayton unit, the system parameters will soon stabilize after fluctuation,

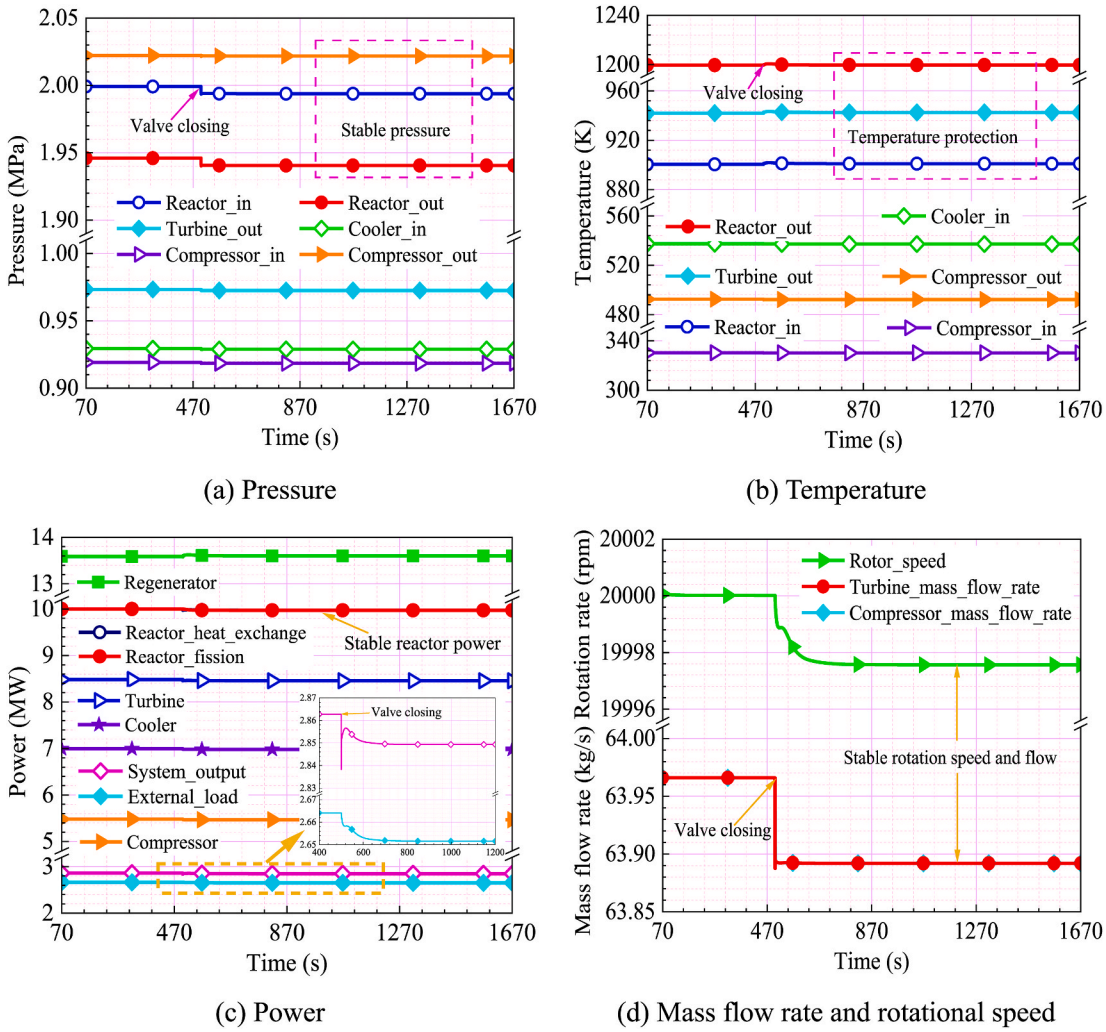


Fig. 28. Transient curve of system operating parameters under the protected condition of the load following protection control.

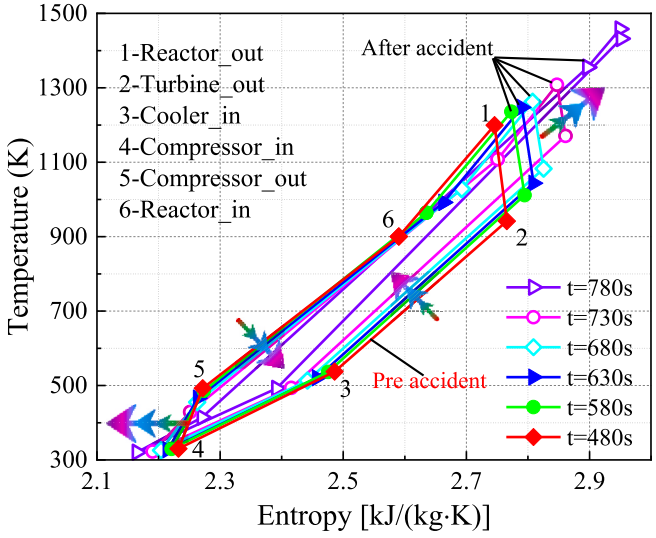


Fig. 29. Temperature-specific entropy (T-s) diagram under the unprotected condition of the step decrease in efficiency of Brayton unit.

and the overall change is small. Realize temperature protection and avoid the risk of ultra-high temperature. Therefore, when the efficiency step down accident of Brayton unit occurs, the system can be stable in another steady state under the load following protection state, and the accident risk is greatly weakened. This also proves that the load following protection strategy can achieve the safety protection of the reactor system in the event of a step down in the efficiency of the Brayton unit.

4.5. Analysis of accident unprotected characteristic parameters

Fig. 33 shows the variation characteristics of key parameters of different accidents under unprotected control, which can be used for accident identification. The unique feature of the external load loss accident is that the rotating speed, reactor power and maximum system pressure have an upward trend. The unique feature of heat sink loss accident is the decrease of cooling water mass flow rate. The unique feature of the Main valve accidental closure accident is the step drop of reactor pressure. The unique feature of the Brayton unit efficiency step decrease accident is the step rise of reactor temperature.

5. Conclusion

The safety characteristic analysis program for helium-xenon cooled reactor system is developed based on Modelica language. Firstly, based on the physical properties of helium-xenon mixed gas, three

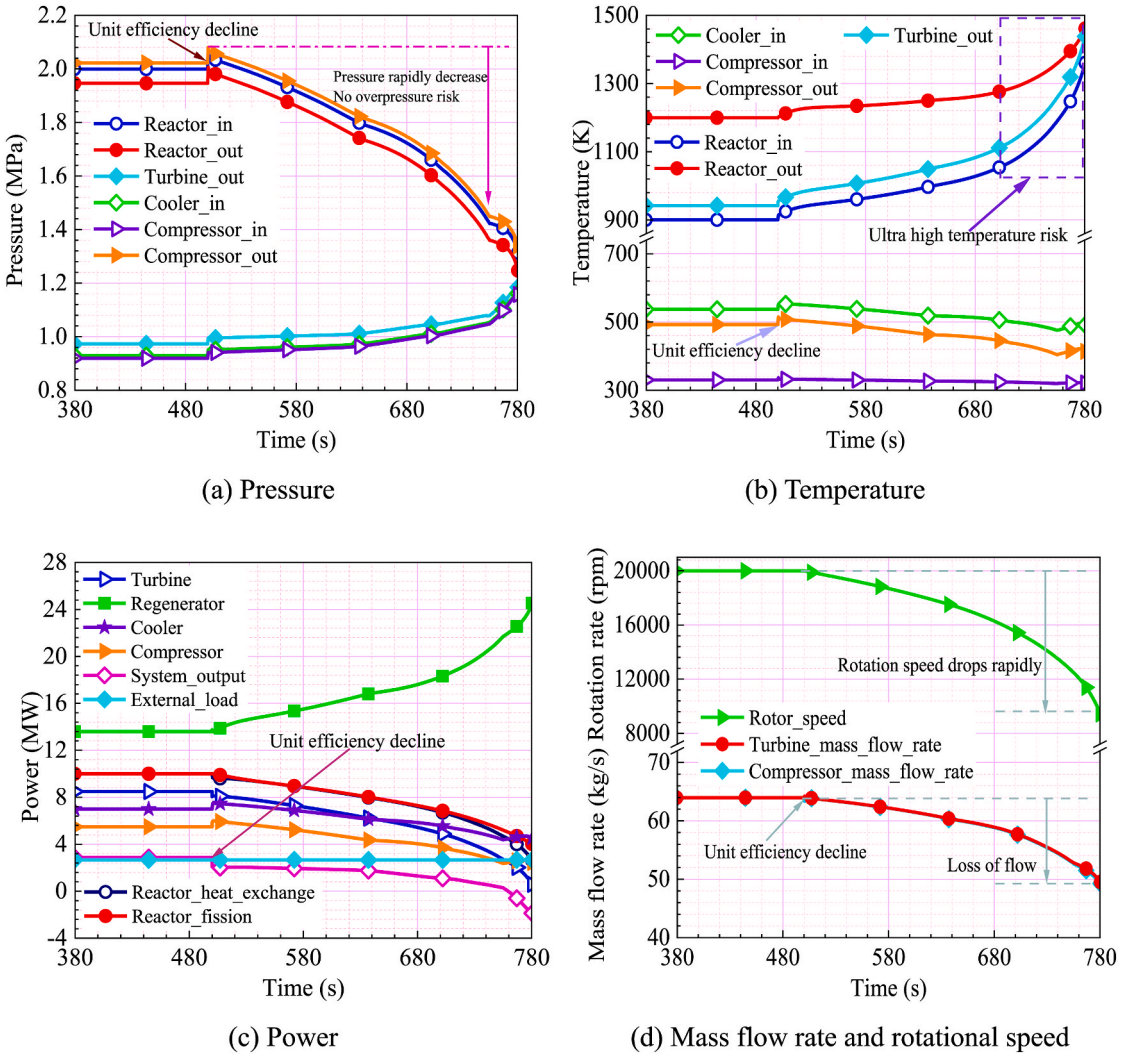


Fig. 30. Transient curve of system operating parameters under the unprotected condition of the step decrease in efficiency of Brayton unit.

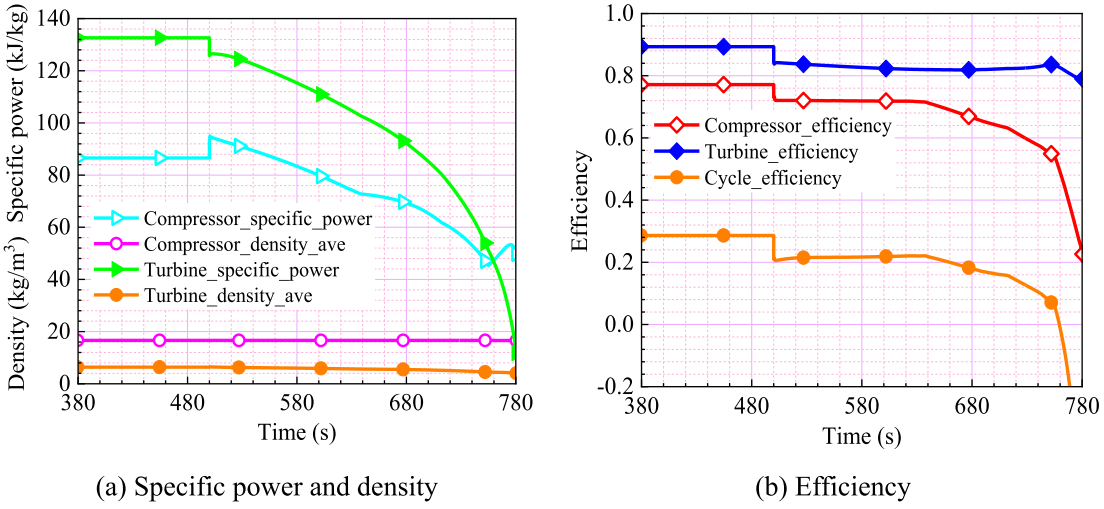


Fig. 31. Transient curve of rotating mechanical parameters under the unprotected condition of the step decrease in efficiency of Brayton unit.

conservation equations, and gas state equation, key equipment models of modular helium-xenon cooled reactor system were established. Then, the transient verification was conducted, and the overall trend was roughly consistent with the literature values, with a maximum relative error of 7.91 %. Finally, the transient conditions of without protection accident are simulated, such as external load loss accident, partial loss of

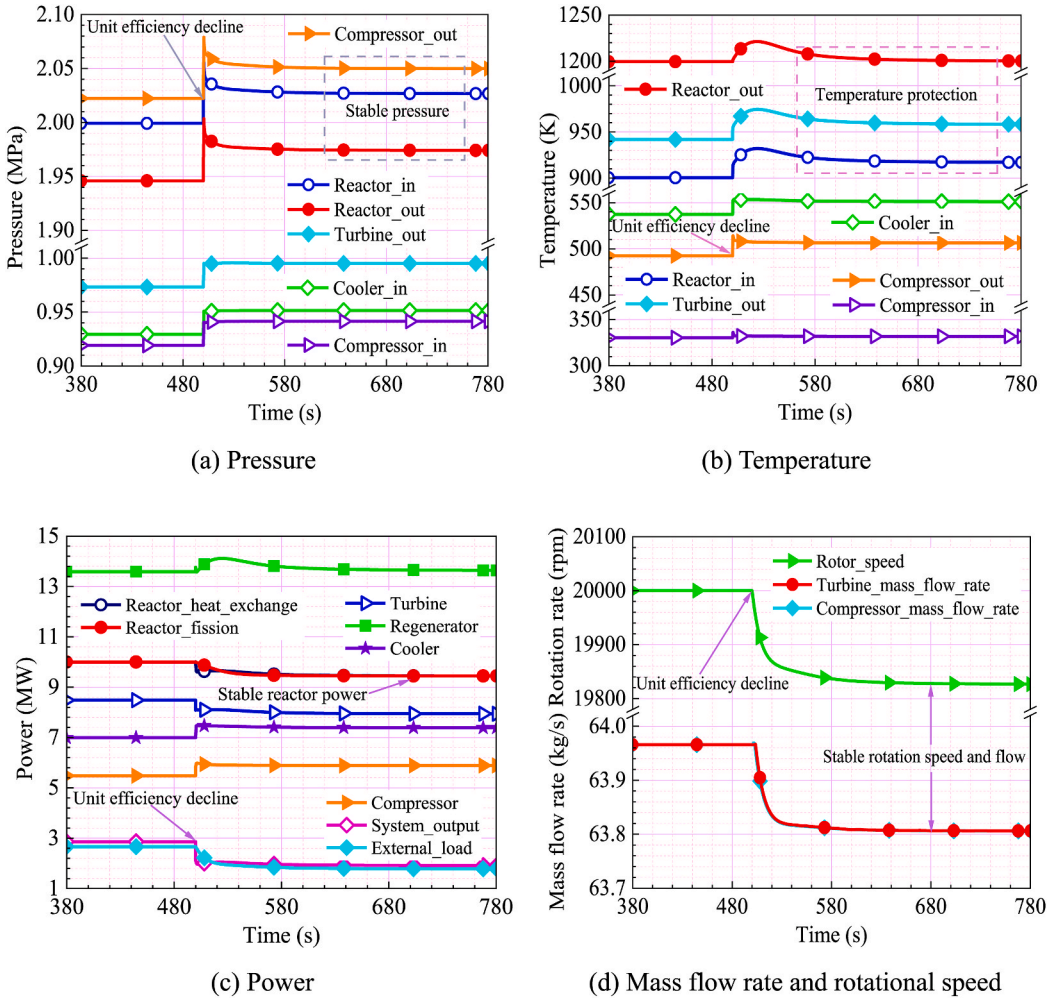


Fig. 32. Transient curve of system operating parameters under the protected condition of the load following protection control.

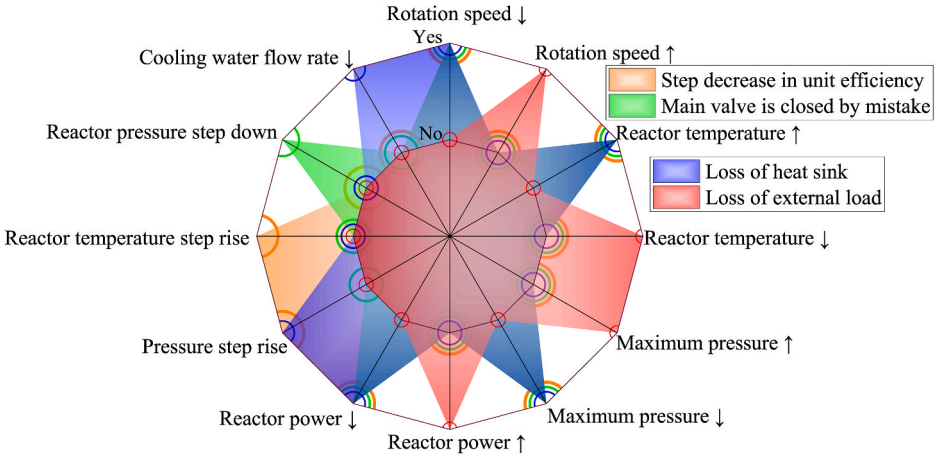


Fig. 33. Comparison diagram of changes in key parameters of each accident.

heat sink accident, main valve accidental closure accident and Brayton unit efficiency step decline accident. According to the no protection characteristics of each accident, the corresponding protection control strategies is proposed, and the transient characteristics of protection control are analyzed. The main conclusions are as follows.

(1) The loss of external load will lead to an increase in rotation speed, system maximum pressure and reactor power, and the rate of increase will increase with the degree of load loss. The introduction of negative reactivity or bypass control can well realize the protection of load loss accident.

(2) The reactor power and outlet temperature under the Introducing negative reactivity protection are 55.46 % and 66.69 % of those

under the bypass control, respectively. In addition, the bypass control has the risk of overpressure. The introduction of negative reactivity protection control is better than bypass control.

(3) The heat sink partial loss, main valve accidental closure and Brayton unit efficiency step decrease have similar unprotected characteristics. Both of them will lead to the rapid decline of rotation speed and mass flow rate, and the risk of reactor temperature exceeding the limit.

(4) The load following protection control can well realize the protection of the heat sink partial loss, main valve accidental closure and Brayton unit efficiency step decrease accident. The rotation speed and mass flow rate are stabilized and the reactor temperature is protected.

In the follow-up study, the system analysis program model needs to be further improved, such as adding more control strategy models and improving the accuracy of equipment models. The verification of the system analysis program still needs to be further carried out and the possibility of experimental verification should be explored. Design more accident transient conditions that the reactor system may encounter, put forward more corresponding protection and control strategies, and continuously optimize and iterate the control strategy scheme. In addition, the helium-xenon Brayton cycle system experimental platform will be built in the follow-up to carry out the following research: verifying the system analysis program, system startup and shutdown transient characteristics, heat source power step or linear change transient characteristics, system pressure boundary break transient characteristics, various typical accident conditions transient characteristics, control strategy selection and comparative research.

CRediT authorship contribution statement

Haoyang Liao: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Xianbo Wang:** Validation, Formal analysis, Data curation. **Lin Xie:** Validation, Formal analysis, Data curation. **Ruibo Lu:** Validation, Formal analysis, Data curation. **Fulong Zhao:** Writing – review & editing, Supervision, Resources. **Sichao Tan:** Writing – review & editing, Supervision, Funding acquisition. **Puzhen Gao:** Writing – review & editing, Supervision, Funding acquisition. **Ruifeng Tian:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

We declare that we have no conflict of interest.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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